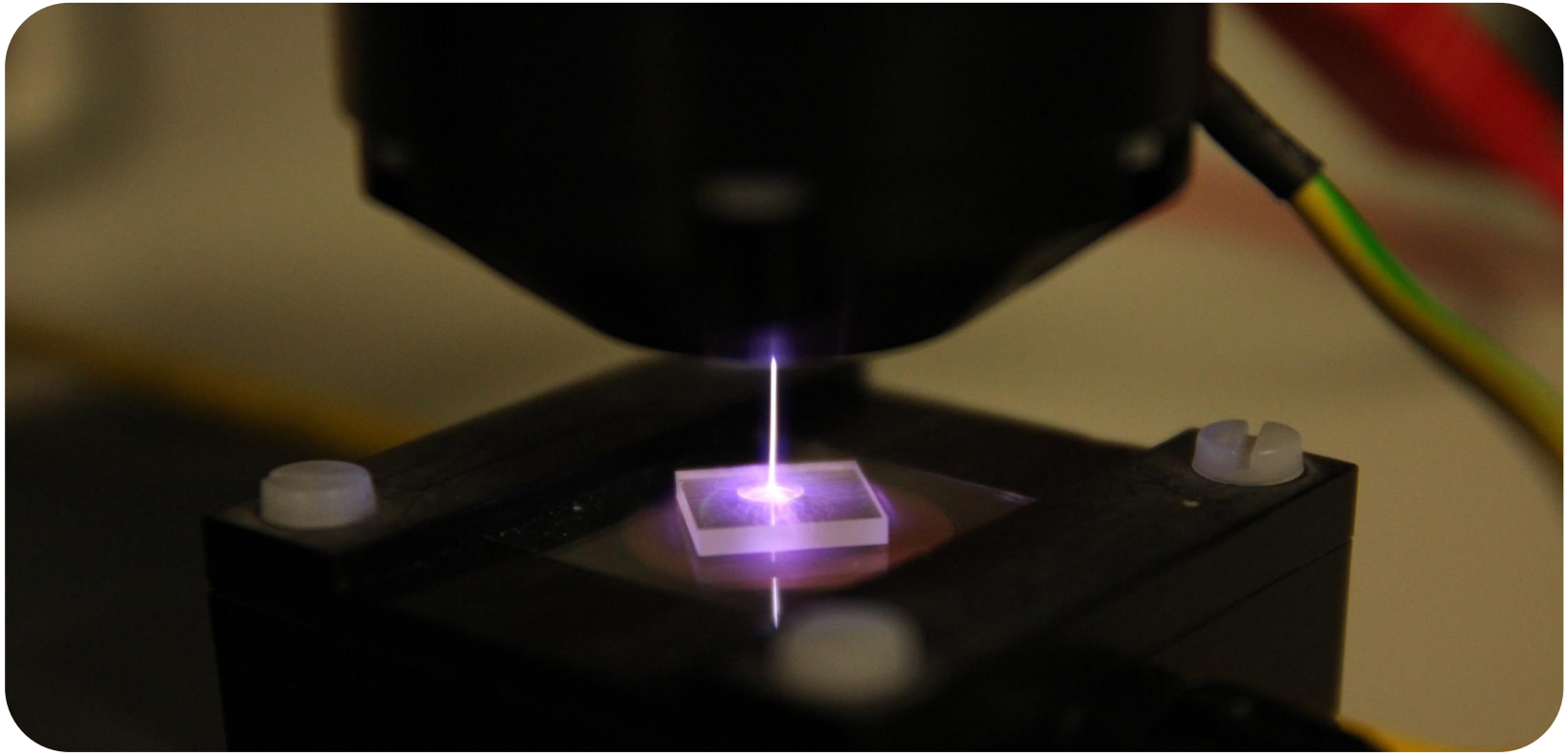
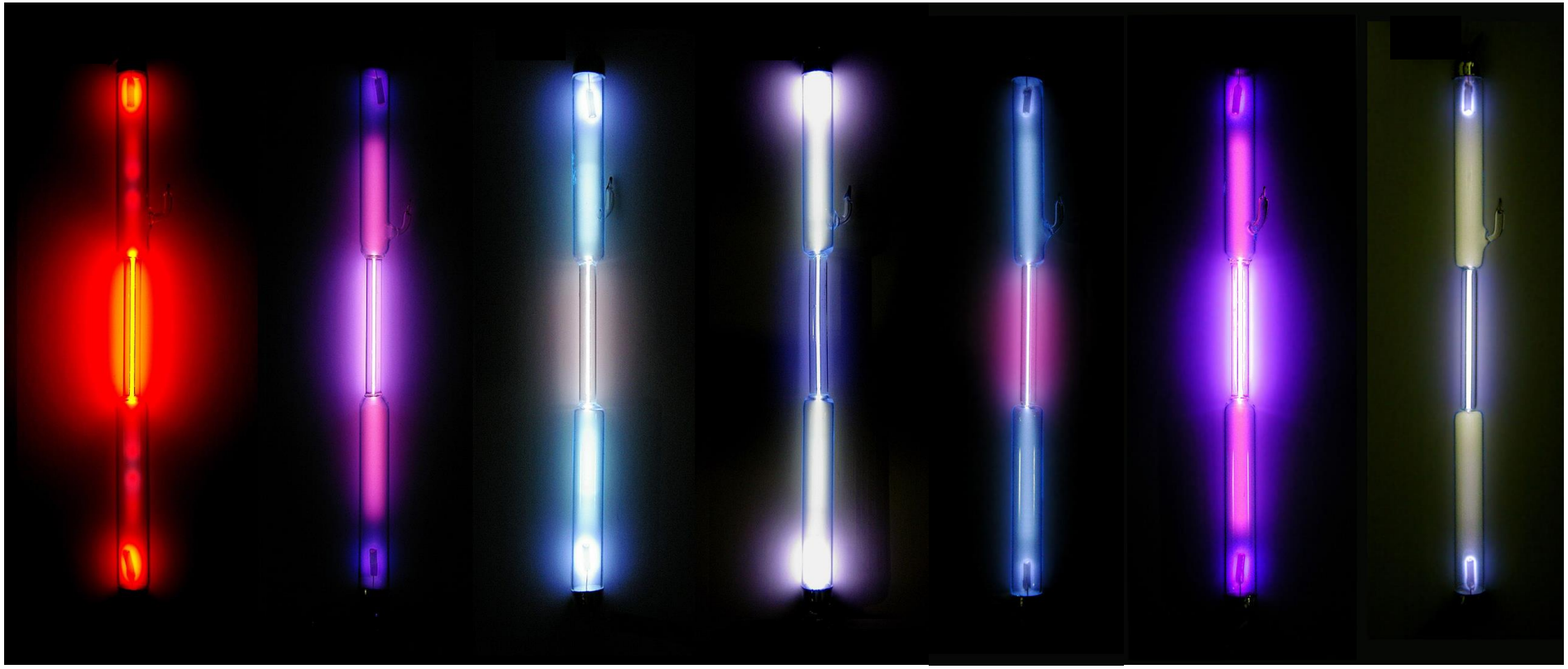
2 Plasma discharges and sources



2.1 Gas discharge plasmas

Gas discharge plasmas

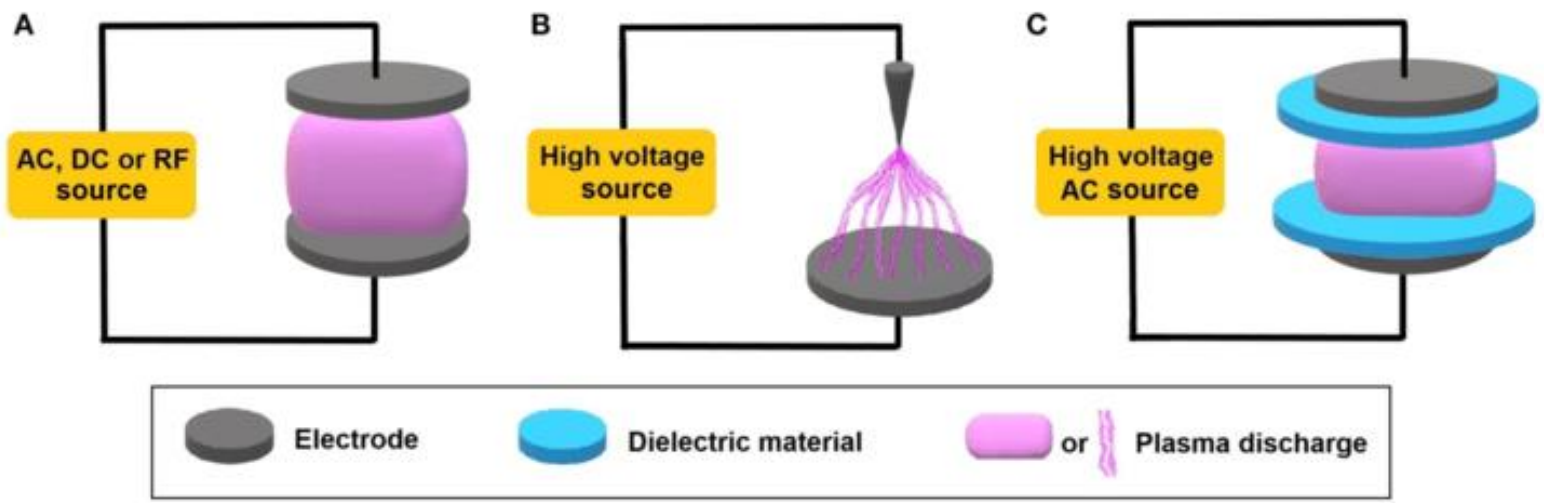
- **Definition**
  - Gas discharge plasma = electrical discharge within a (process) gas



Neon (Ne)	Argon (Ar)	Krypton (Kr)	Xenon (Xe)	Hydrogen (H_2)	Nitrogen (N_2)	Oxygen (O_2)

Gas discharge plasmas

- **Influencing factors**
  - Design of discharge geometry
    - Basic principle:
      - Arc discharge, glow discharge, dielectric barrier discharge (DBD) etc.
    - Type:
      - Direct or indirect
    - Setup:
      - Plate-to-plate (A and C), needle-to-needle, needle to plate (B), needle to sphere, etc.

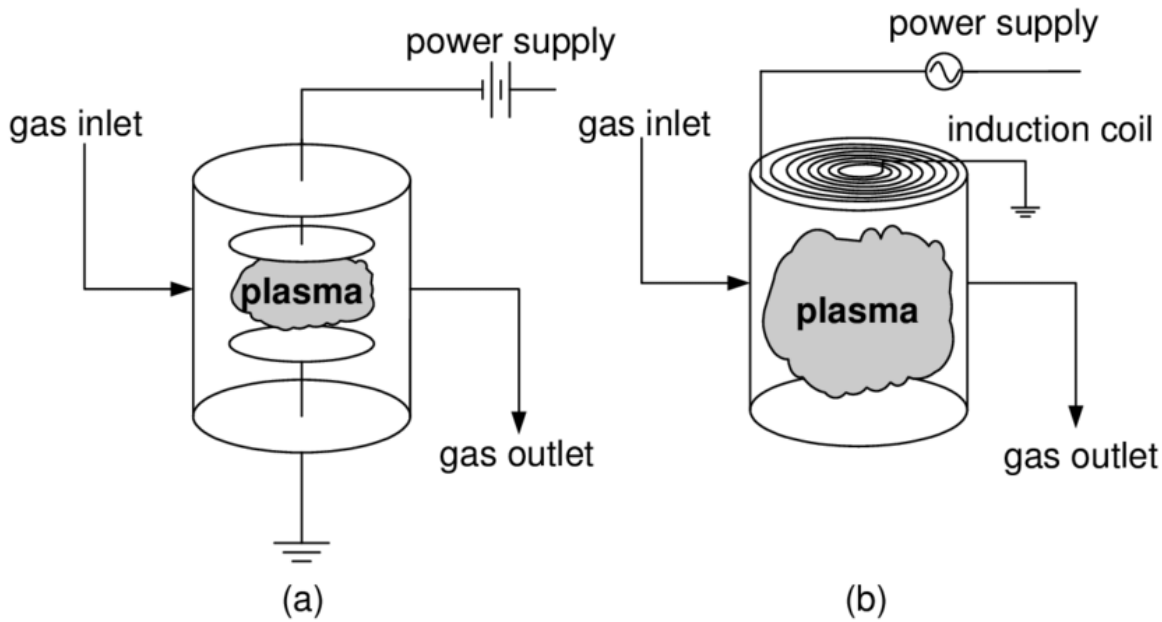


DOI:10.3389/fmech.2020.00042

Gas discharge plasmas

- **Influencing factors**
  - Type of excitation
    - Radio frequency (RF)
    - Alternating current (AC), incl. pulse duration and shape
    - Direct current (DC)
    - High-voltage (HV), low voltage (LV)
    - etc.

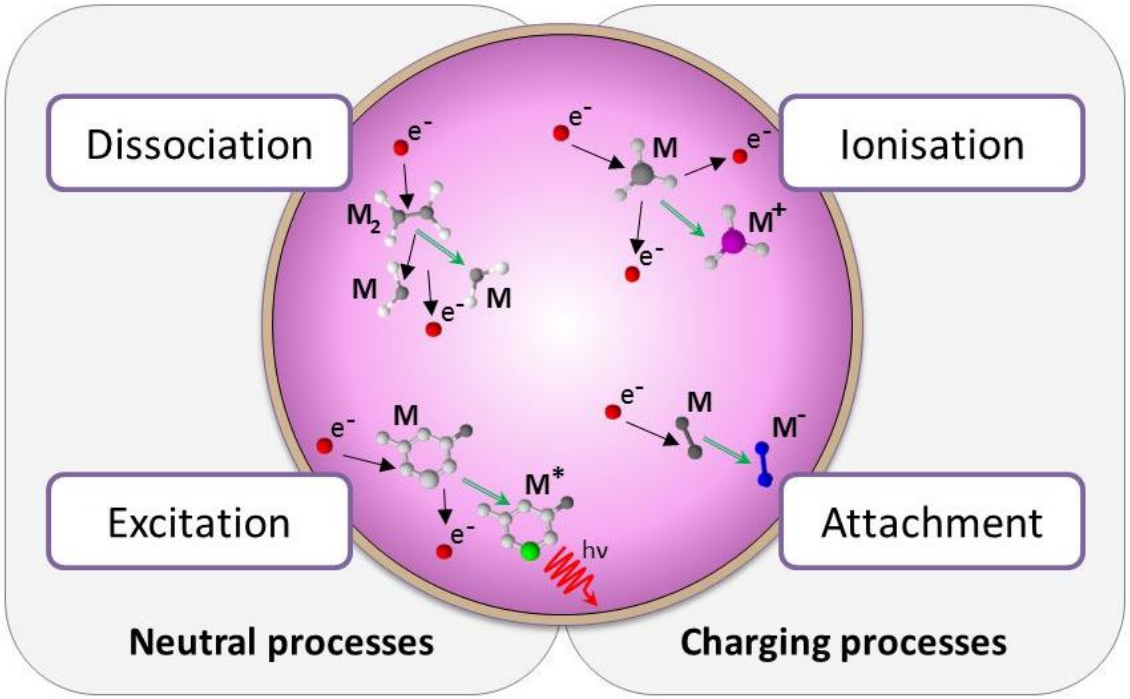
  - Supply of electrical energy
    a) Capacitively
    b) Inductively



David Humbird, Thesis, University of California, Berkeley, Spring 2004

Gas discharge plasmas

- **Influencing factors**
  - Interaction of particles
    - Concentration gradients → diffusion
    - Collisions, leading to
      - Dissociation
      - Ionisation
      - Attachment
      - Excitation
    - etc.



Gas discharge plasmas

- **Influencing factors**
  - Electrical and magnetic fields
    - → drift
  - Formation of local charges, fields and potentials
    - Sheath-effect
  - Magnetic fields induced by electrical currents within the plasma
    - Deflection of charge carriers (see “Larmor-radius” in 1.2.5)
  - *…to be continued…*

Gas discharge plasmas

- **Possible loss processes**
  - Heat transfer
    - Convection of plasma volume and – eventually – change in essential type of plasma
  - Photon emission
    - (an advantageous loss process, BTW,
      → plasma diagnostics via emission spectroscopy)
  - Losses of charge carriers
  - Chemical modification of plasma composition
    - …due to collisions, dissociation, ionisation, attachment

→ **Gas discharge plasmas are quite complex phenomena where several time-dependent terms and mechanisms are involved.**

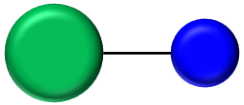
2.1 Gas discharge plasmas

2.1.1 Isothermal and equilibrium plasmas

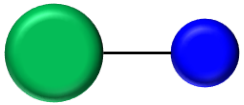
Isothermal and equilibrium plasmas

- **Definition**
  - Thermal plasmas in thermodynamic equilibrium
    → The degrees of freedom of all particles feature the same energy/velocity/temperature

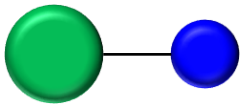
  - ‘Degrees of freedom’:
    - Translation = linear motion in different directions in space

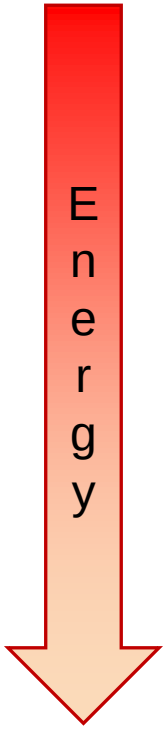


    - Rotation = rotation (of molecules) around different axes



    - Vibration
      = vibrations around a certain position in different directions in space





Isothermal and equilibrium plasmas

- **Characterisation**
  - Stationary thermal plasmas are calculated and characterised by statistic thermodynamics applying the
    - Maxwell equation (→ velocities), the
    - Boltzmann equation (→ population densities), and the
    - Saha equation (→ composition).
  - Here, all occupied energy levels E and the corresponding statistic weights g, the partition functions Q, the ionisation and dissociation energies and the energies of all chemical reactions within the plasmas need to be known.

Velocities: $$f(\upsilon) = 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} \upsilon^2 \; e^{-m\upsilon^2/2kT}$$ *Maxwell*

Population densities: $$n_i = n \frac{g_i}{Q} e^{-E_i/kT}$$ *Boltzmann*

Composition: $$\frac{n_A n_B}{n_{AB}} = \frac{(2\pi \mu kT)^{3/2}}{h^3} \frac{Q_A Q_B}{Q_{AB}} e^{-E_{AB}/kT}$$ *Saha*

* In a stationary plasma, all eventually effluent particles are continuously recouped.
Example: plasma jet or plasma torch

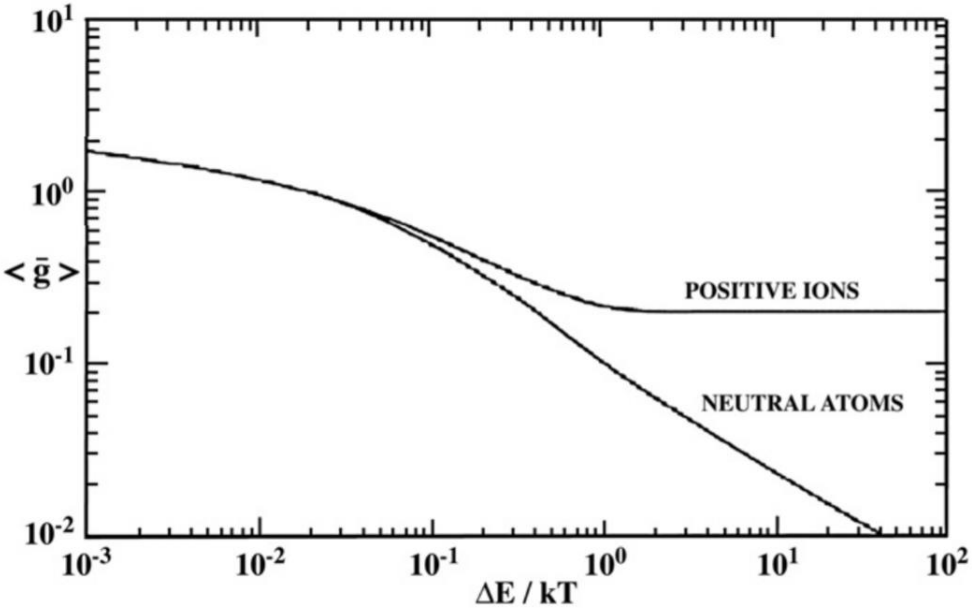
Isothermal and equilibrium plasmas

- **The McWhirter criterion**
  - For the characterisation of isothermal plasmas, the condition for local thermal equilibrium, aka LTE criterion or McWhirter criterion has to be fulfilled:

$$n_e > \frac{2.74 \times 10^{13}}{\langle \bar{g} \rangle} \sqrt{kT} \left(\Delta E_{\max} \right)^3$$

For most elements: $n_e \geq 10^{16}$ cm^{-3}

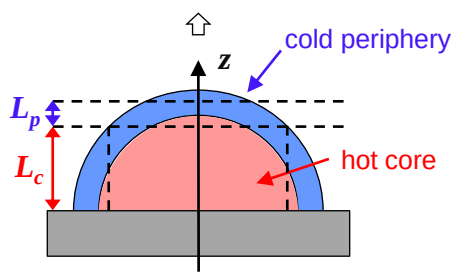
Here, ⟨ḡ⟩ is the dimensionless Gaunt factor (or Kramers-Gaunt factor), used for multiplicative correction to the continuous absorption or emission results when calculated using classical physics techniques

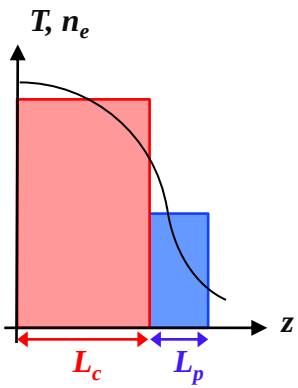


Spectrochimica Acta Part B 65 (2010) 86–95

Isothermal and equilibrium plasmas

- **Restrictions**
  - Note: isothermy not strictly valid for plasmas in electric fields (i.e., any technical plasma) since
    - Light electrons are more accelerated than heavier ions – and thus faster, energy-richer, and 'hotter'
    - Peripheral zone of plasma is usually colder than core (especially for plasmas in contact with surrounding atmosphere, e.g. atmospheric pressure plasmas or laser-ignited plasmas, see example below)





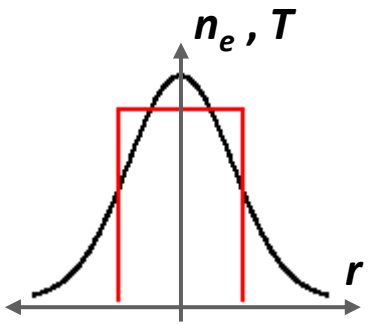
	Plasma core			Peripheral zone		
t (ns)	T (K)	n_e (cm^{-3})	L	T (K)	n_e (cm^{-3})	L
500	12,300	2.7×10^{17}	0.60	8600	6×10^{16}	0.12
7000	6200	2.7×10^{16}	0.85	4800	8×10^{15}	0.20

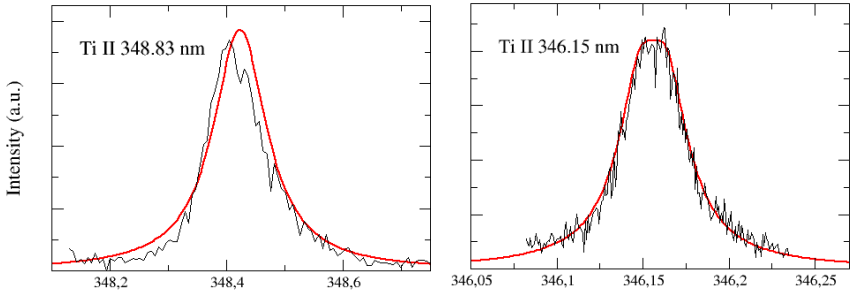
Isothermal and equilibrium plasmas

- **Restrictions**
  - Peripheral zone of plasma is usually colder than core
  - → to be considered during calculation/modelling of plasmas, e.g. plasma emission:

uniform plasma:

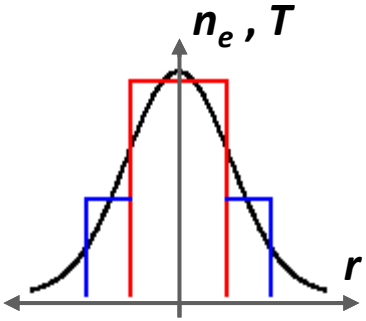
$$B = U\left(1 - e^{-\alpha L}\right)$$

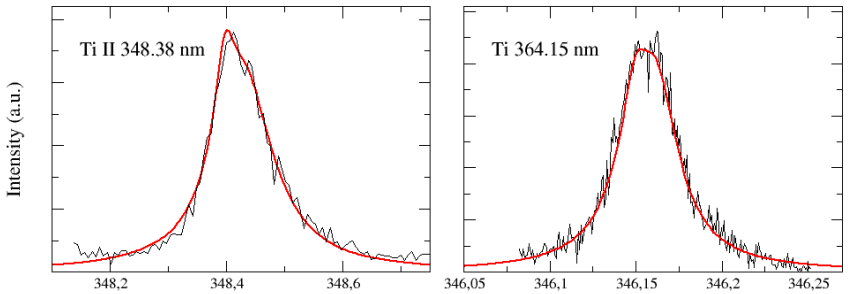




non-uniform plasma (2 zones):

$$B = U_C\left(1 - e^{-\alpha_C L_C}\right)e^{-\alpha_P L_P} + U_P\left(1 - e^{-\alpha_P L_P}\right)$$





B: spectral radiance, *U*: spectral blackbody radiance

Isothermal and equilibrium plasmas

- **Restrictions**
  - Calculation of plasma emission (here: for uniform plasmas) using an approach analogue to the Lambert-Beer-law:

$$B = U\left(1 - e^{-\alpha L}\right)$$

where

$$\alpha(\lambda) = \pi r_0 \lambda^2 f_{lu} n_l P(\lambda) \left[1 - \exp\left(-\frac{hc}{\lambda kT}\right)\right]$$

B: spectral radiance
U: spectral blackbody radiance
α: absorption coefficient
λ: wavelength
r_0: classical electron radius
f_{lu}: absorption oscillator strength
n_l: lower level population density of the particular transition
P(λ): normalised line profile
c: vacuum light velocity

Isothermal and equilibrium plasmas

???

- **Exercise 2.1**
  - The spectral radiance of a blackbody at a temperature of 16000K amounts to $1.88102 \cdot 10^9$ $W \cdot m^{-2} \cdot sr^{-1} \cdot \mu m^{-1}$ at a wavelength of 350 nm. Determine the spectral radiance of the plasma at this wavelength assuming…
    a) …a uniform plasma in LTE with a temperature of T = 16000K, an absorption coefficient of α = 0.005/cm and a length of L = 12 mm and
    b) …a non uniform plasma consisting of two zones, i.e., a core (index c) and a peripheral zone (index p) where
       - T_c = 16000K, α_c = 0.008/cm and L_c = 6 mm and
       - T_p = 3000K, α_p = 0.002/cm and L_p = 6 mm and

  - Practice tip: Blackbody Calculator
    https://www.spectralcalc.com/blackbody_calculator/blackbody.php

2.1 Gas discharge plasmas

2.1.2 Non-isothermal and non-equilibrium plasmas

Non-isothermal and non-equilibrium plasmas

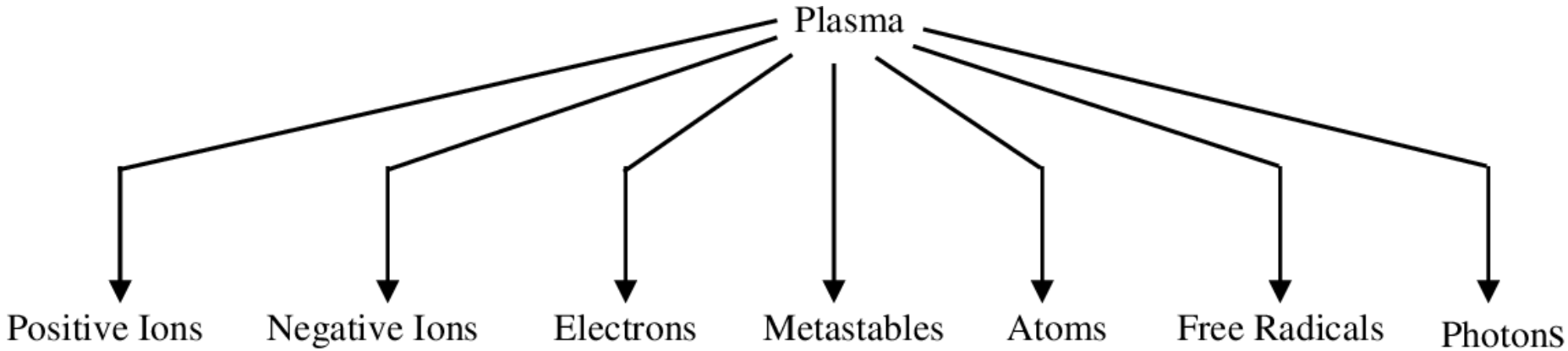
- **Definition**
  - Non-isothermal plasmas
    → The degrees of freedom of all particles DO NOT feature the same energy/velocity/temperature

Example: High electron temperature + low ion/gas temperature = ‘cold’ plasma

Non-isothermal and non-equilibrium plasmas

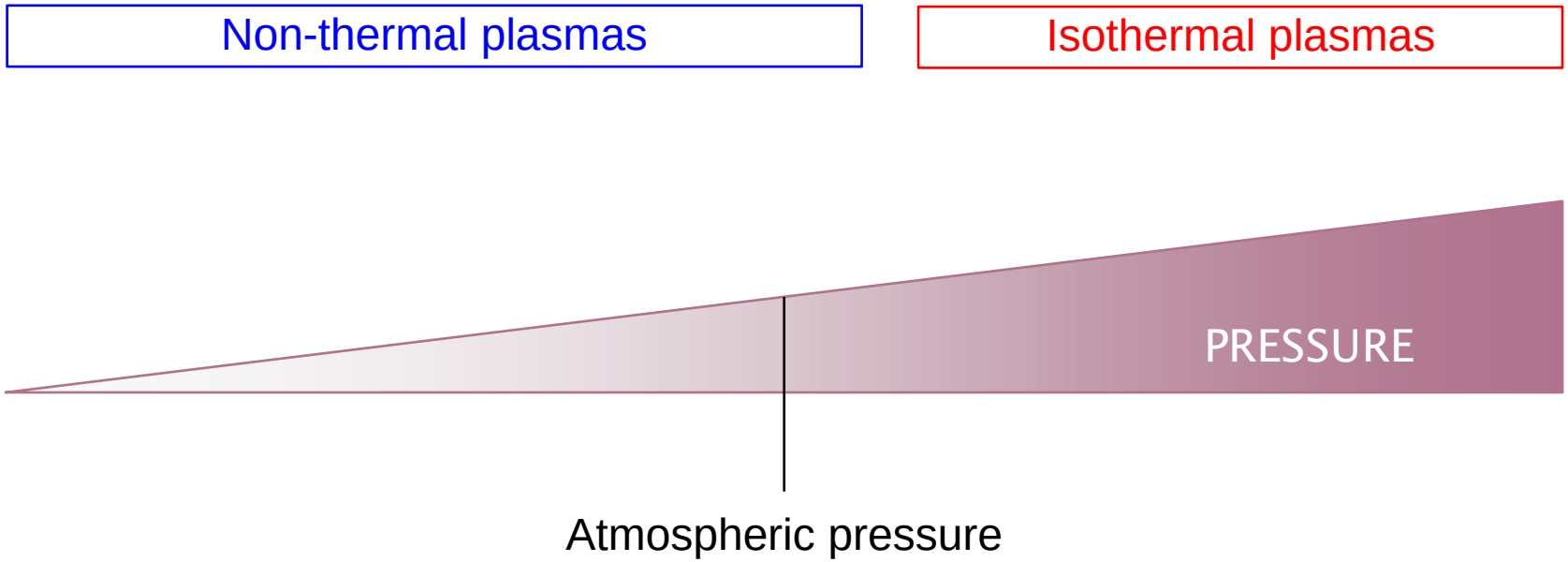
- **Characterisation**
  - Modelling on the basis of the calculation of density and energy distribution of the electron gas via Boltzmann equation
    → Ionisation rate can be approximated via measured ionisation coefficient of the gas

  - The density of excitation of energy levels can merely be estimated if all essential excitation processes (dissociation, photo emission, absorption…) are known/measured
    → More data required in comparison to LTE plasma



Non-isothermal and non-equilibrium plasmas

- **Characteristics and comparison**
  - Isothermal plasmas are formed within stationary gas discharges at pressures > 1013 hPa where many collisions of gas particles occur
  - Stationary non-isothermal plasmas occur at atmospheric pressure or lower pressures



Non-isothermal and non-equilibrium plasmas

- **Effects leading to deviations from thermodynamic equilibrium:**

  1. Statistic sum of all collisions and energy absorption in electric field results in electron energy distribution function EEDF

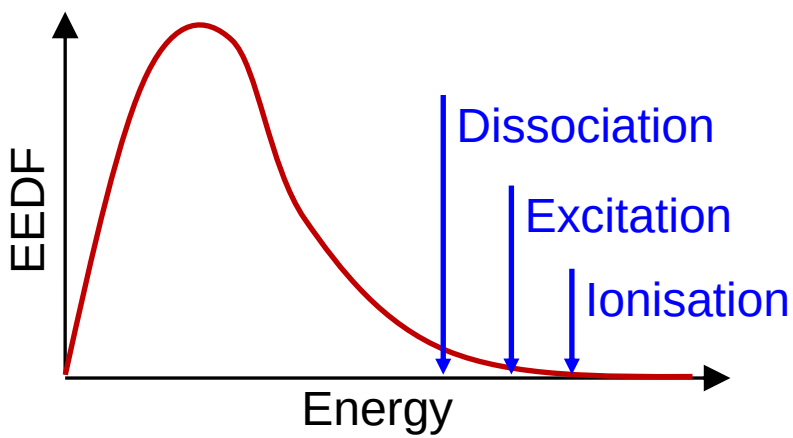
  2. Mean electron energy is higher than mean excitation energy of heavier particles
     ($E_{electron} > E_{excitation}$)

  3. Mean electron temperature is higher than translation temperature (= ‘true gas temperature’) of heavy particles
     ($E_{electron} > E_{gas}$)

  4. Vibrational temperature is higher than rotational temperature
     ($E_{vib} > E_{rot}$)

Non-isothermal and non-equilibrium plasmas

- **The electron energy distribution function EEDF**
  - Electron energy distribution function EEDF ≠ Maxwell-distribution, if electron density is low in comparison to gas density
    → Collisions between electrons rare compared to inelastic collisions between electrons and heavier particles
  - Inelastic collisions → transfer of excitation energy from electron to collision partner
    → Reduction of free electrons of high energy
    → Distribution of electron energy



Non-isothermal and non-equilibrium plasmas

- **The electron energy distribution function EEDF**
  - *Excursus*: On the kinetic energy of electrons:

When applying a voltage to a gas-filled gap, charge carriers can be accelerated and effect the ionization of the gas. Such ionization can further be increased by electron avalanches for sufficient voltages. However, the existence of initial electrons, for example caused by cosmic radiation or the photo effect, is a precondition for this mechanism. The electric field E is generally given by

$$E = -\frac{d\phi}{dx}. \qquad 1$$

By replacing the differentially small distance dx by the mean free path λ_m and the change in electrical potential $d\phi$ by the difference of potential U_{λ_m}, which is passed by the electron for this distance, E can be expressed as

$$E = -\frac{U_{\lambda_m}}{\lambda_m}. \qquad 2$$

A homogenous electric field is assumed for simplification. The following relationship results when taking the kinetic energy, which is absorbed by the electron when passing the distance λ_m within the electric field, into account:

$$E_{kin} = \frac{m_e \cdot v^2}{2} = e_0 \cdot U_{\lambda_m} = e_0 \cdot E \cdot \lambda_m. \qquad 3$$

Non-isothermal and non-equilibrium plasmas

- **The electron energy distribution function EEDF**
  - EEDF can be calculated via kinetic interrelationships and equations, where all loss mechanisms and terms need to be considered
  - Description via
    - Maxwell function[1],

$$f(\epsilon) = \varphi^{-3/2}\beta_1 \exp\left(-\frac{\epsilon\beta_2}{\varphi}\right) \qquad \beta_1 = \Gamma(5/2)^{3/2}\Gamma(3/2)^{-5/2},\ \beta_2 = \Gamma(5/2)\Gamma(3/2)^{-1}$$

    - Druyvesteyn function[2],

$$f(\epsilon) = \varphi^{-3/2}\beta_1 \exp\left(-\left(\frac{\epsilon\beta_2}{\varphi}\right)^2\right) \qquad \beta_1 = \Gamma(5/4)^{3/2}\Gamma(3/4)^{-5/2},\ \beta_2 = \Gamma(5/4)\Gamma(3/4)^{-1}$$

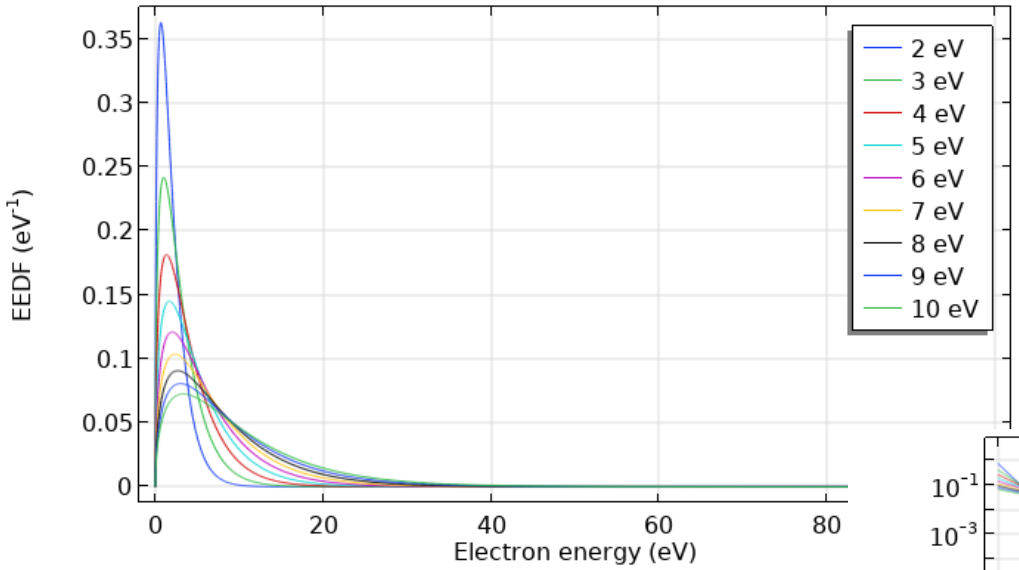
ε = electron energy, φ = mean electron energy, and Γ = incomplete Gamma function (e.g. Γ(3/4) = 1.2254)

[1]using a constant collision frequency

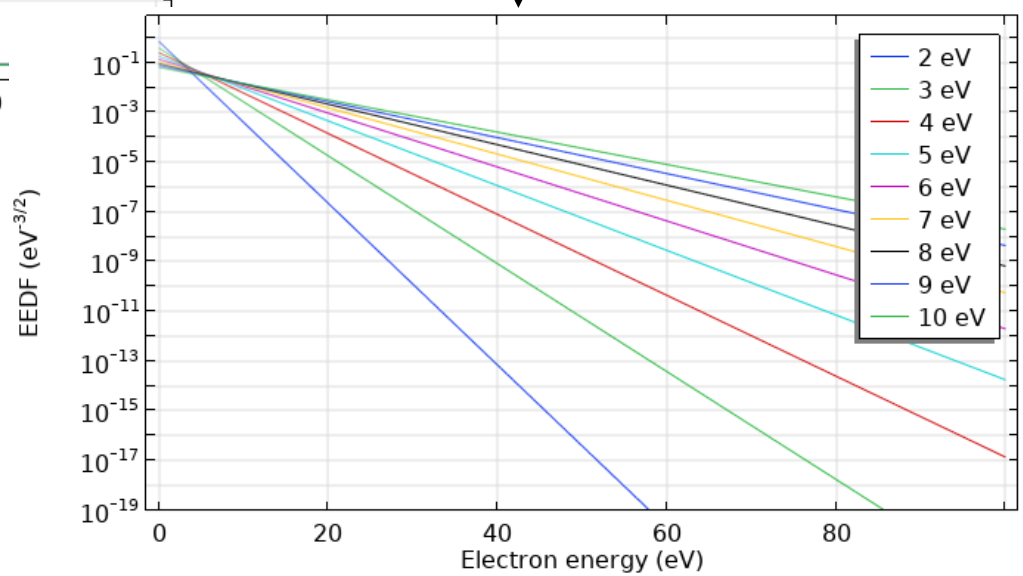
[2]assuming a constant, electron-energy-independent cross section describing loss/gain of energy via collisions with background gas

Non-isothermal and non-equilibrium plasmas

- **The electron energy distribution function EEDF**
  - *Example: Maxwellian EEDF*



Division by √ε gives commonly used type, aka electron energy probability function (EEPF)



https://www.comsol.com/blogs/electron-energy-distribution-function/

Non-isothermal and non-equilibrium plasmas

- **The electron energy distribution function EEDF**
  - EEDF can also be determined by
    - Intermediate function, combining Maxwell and Druyvesteyn function:

      $$f(\epsilon) = \varphi^{-3/2}\beta_1 \exp\left(-\left(\frac{\epsilon\beta_2}{\varphi}\right)^g\right) \quad \beta_1 = \Gamma(5/2g)^{3/2}\Gamma(3/2g)^{-5/2},\ \beta_2 = \Gamma(5/2g)\Gamma(3/2g)^{-1}$$

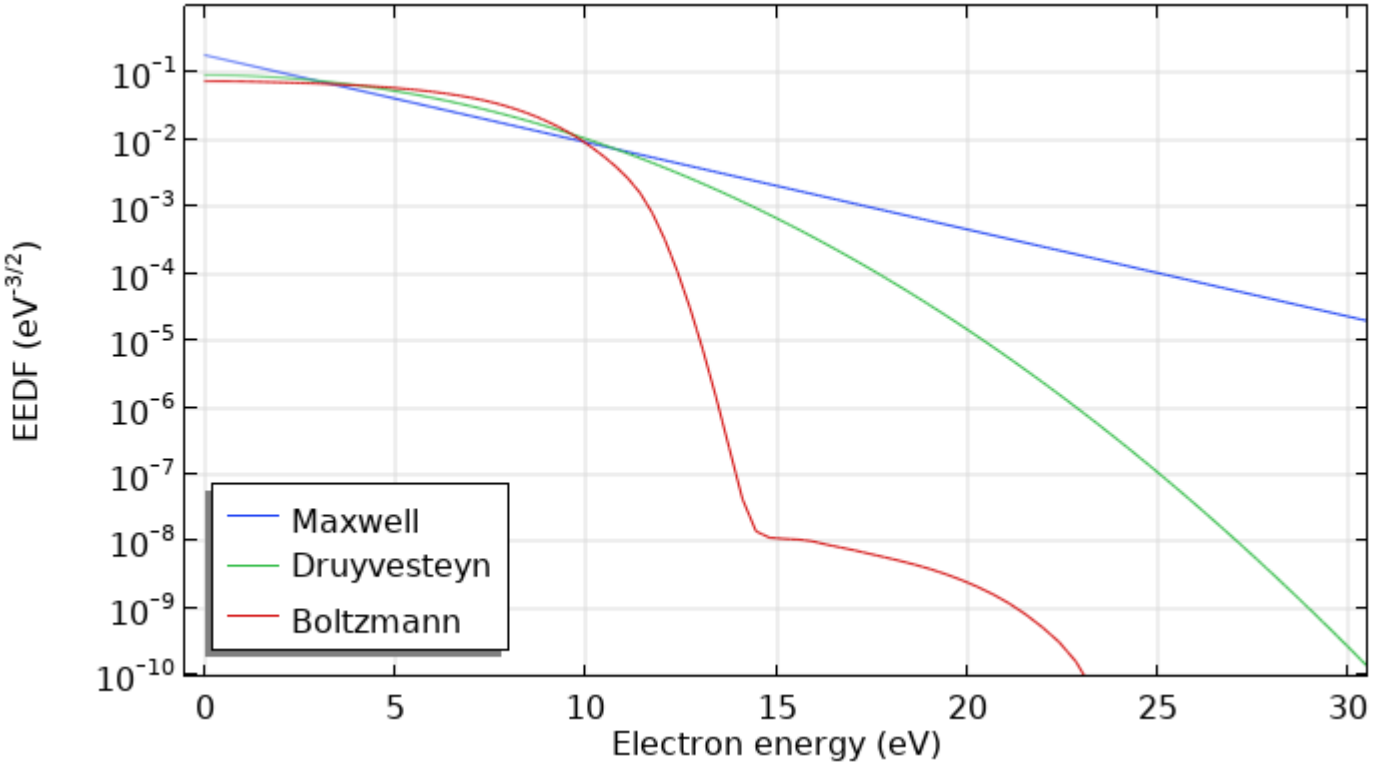
      g = specific factor (= 1 for Maxwell distribution and 2 for Druyvesteyn distribution)
    - or by solving the Boltzmann equation:

      $$\frac{\partial f}{\partial t} + \mathbf{v}\cdot\nabla f - \frac{e}{m}(\mathbf{E}\cdot\nabla_{\mathbf{v}} f) = C[f]$$

    → most accurate method due to
      - Introduction and consideration of anisotropic perturbation and
      - Additional consideration of
        - Energy gain of electrons caused by the electric field *E* and
        - Energy loss of electrons resulting from collisions within a specific gas or gas mixture

Non-isothermal and non-equilibrium plasmas

- **The electron energy distribution function EEDF**
  - Maxwell vs. Druyvesteyn vs. Boltzmann:



  - Electrons in ‘thermodynamic equilibrium’ → Maxwellian distribution function (only valid for high degrees of ionisation)
  - ‘Real’ technical plasma → Boltzmann equation-based EEDF

https://www.comsol.com/blogs/electron-energy-distribution-function/

Non-isothermal and non-equilibrium plasmas

- $E_{electron} > E_{excitation}$
  - Mean electron energy is higher than mean excitation energy of heavier particles
    → Within a plasma, higher energy levels N_j are excited from the ground state N_0 by fast electrons e*:

$$N_0 + e^* \longrightarrow N_j + e$$

  - Excitation probability from state 0 to j depends on excitation cross section σ, which is a function of the isotropic electron velocity v_e:

$$\sigma_{0\to j}(v_e)$$

For an electron density n_e and a population density at the ground state n_0, the excitation rate is:

$$\frac{\mathrm{d}n_j}{\mathrm{d}t} = n_0 n_e X_{0\to j}; \quad X_{0\to j} = \int_0^\infty \sigma_{0\to j}(v_e)\mathrm{d}v_e$$

Non-isothermal and non-equilibrium plasmas

- $E_{electron} > E_{excitation}$
  - Excitation rate within plasma
    → Extremely non-uniform population of energy levels due to different excitation cross sections
    → Boltzmann-distribution is not existent before backscatter process:

$$N_j + e \longrightarrow N_0 + e^*$$

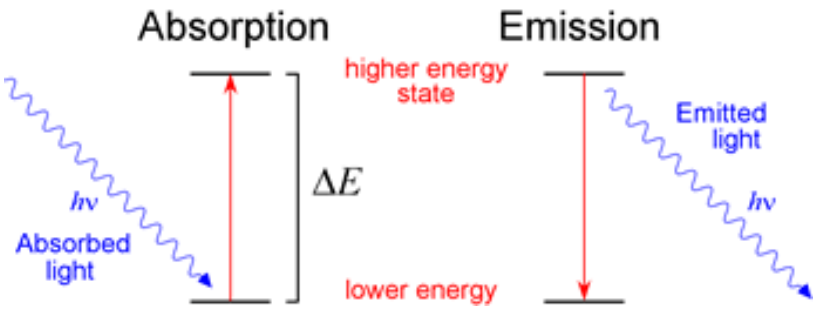
  - Backscatter processes (= super-elastic collisions)
    → Formation of detailed equilibrium due to spontaneous emission of photons where the emission rate is:

$$\boxed{\text{Photon emission rate } \frac{\mathrm{d}n_j}{\mathrm{d}t}} = \boxed{-A_{ji} \cdot n_j \text{ Photon generation rate}}$$

A: Einstein-coefficient of spontaneous emission for the particular transition, here: j to i

Non-isothermal and non-equilibrium plasmas

- $E_{electron} > E_{excitation}$
  - Inverse process of photon emission is photon absorption within lower i-levels, leading to excitation of higher j-levels (= population inversion)



Photon density rate = photon generation rate – photon absorption rate:

$$\frac{\mathrm{d}q_{ji}}{\mathrm{d}t} = A_{ji} n_j - Q_{ij} q_{ji} c \cdot n_i$$

with the photon absorption cross section:

$$Q_{ij} = \frac{\lambda^2}{8\pi} A_{ij}$$

Non-isothermal and non-equilibrium plasmas

- $E_{electron} > E_{exciation}$
  - Plasma features a certain photon absorption
  - Optically thick plasma, if photons are absorbed after short absorption distance s_{abs}, depending on the photon absorption cross section Q_{ij}:

$$s_{abs} = 1/Q_{ij} n_i$$

→ Optically thick plasmas, if absorption distance is much smaller than distance to edge of plasma ($s_{abs} << L$)

  - Especially of interest in case of
    - Laser materials processing
      (plasma shielding of incident laser irradiation)
    - Laser spectroscopy
      (self absorption of emission lines)
    - Development of laser sources
      (plasma-absorbers as optical switches)

Non-isothermal and non-equilibrium plasmas

- $E_{electron} > E_{gas}$
  - Mean electron temperature is higher than translation temperature (= ‘true gas temperature’) of heavy particles, if electric fields are existent within the plasma
    - *Prime example: Dielectric barrier discharge*
      Here, the plasma state is maintained thanks to a fast alternation (some kHz) of the direction of the field vector, i.e., the polarity of the electric field
      → Fast electrons and (nearly) static ions and neutrals
      → Non-equilibrium
      (Equilibrium, if difference in temperature of charge carriers and particles small with respect to mean plasma temperature)

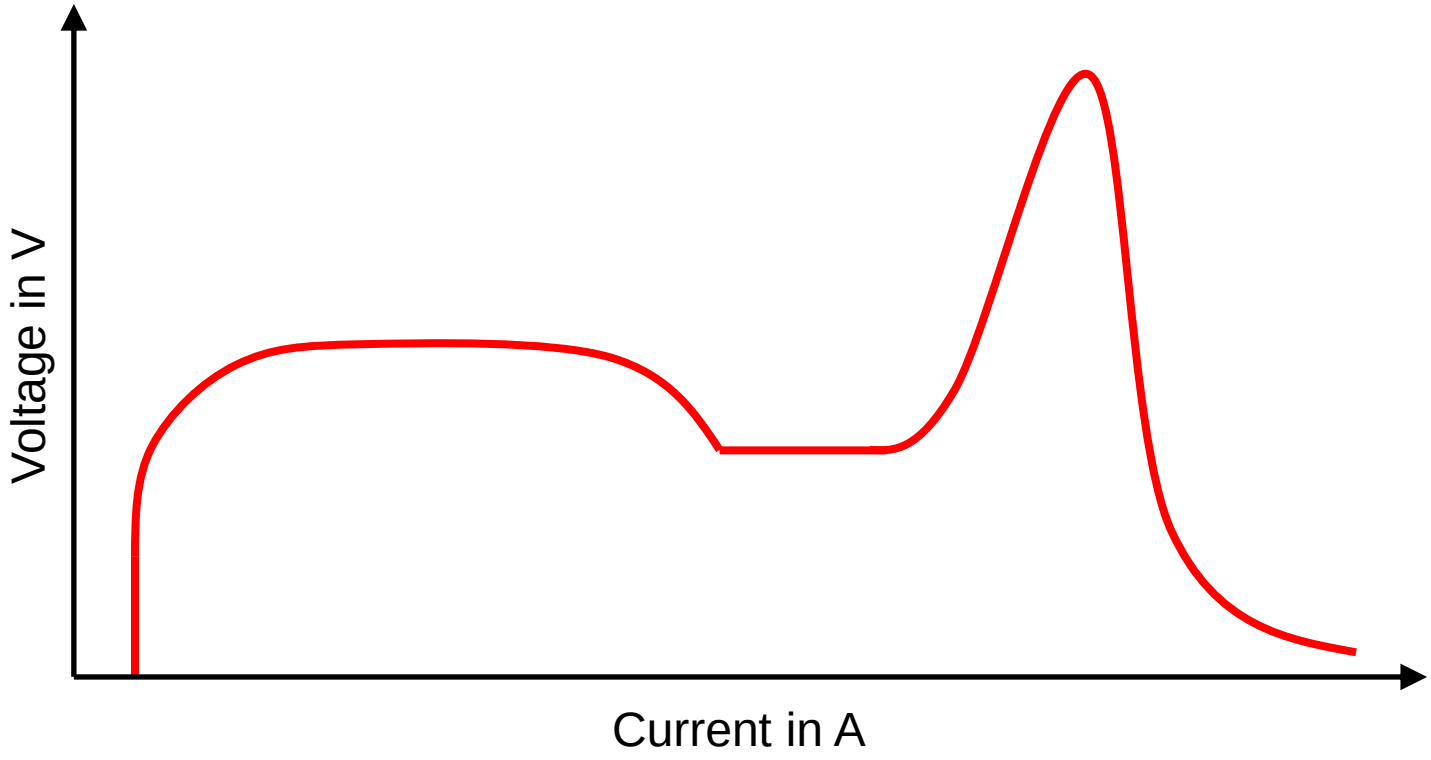
Non-isothermal and non-equilibrium plasmas

- $\boldsymbol{E_{vib} > E_{rot}}$
  - Vibrational temperature is higher than rotational temperature since excitation cross sections of vibrational states are usually higher than for rotational states
  - Quite comparable magnitude of energy required for translation and rotation of atoms and molecules
    → Differences in energy are usually balanced after some collisions (0.1 ns at 1013 hPa and 300K)
    → $E_{trans} \approx E_{rot} \approx E_{gas}$ or
    $\boldsymbol{T_{trans} \approx T_{rot} \approx T_{gas}}$
  - Transfer of energy between vibration and translation notably slower due to significantly higher differences in energy
    → $E_{vib} > E_{rot} \approx E_{gas}$ or
    $\boldsymbol{T_{vib} > T_{rot} \approx T_{gas}}$

2.2 DC discharges

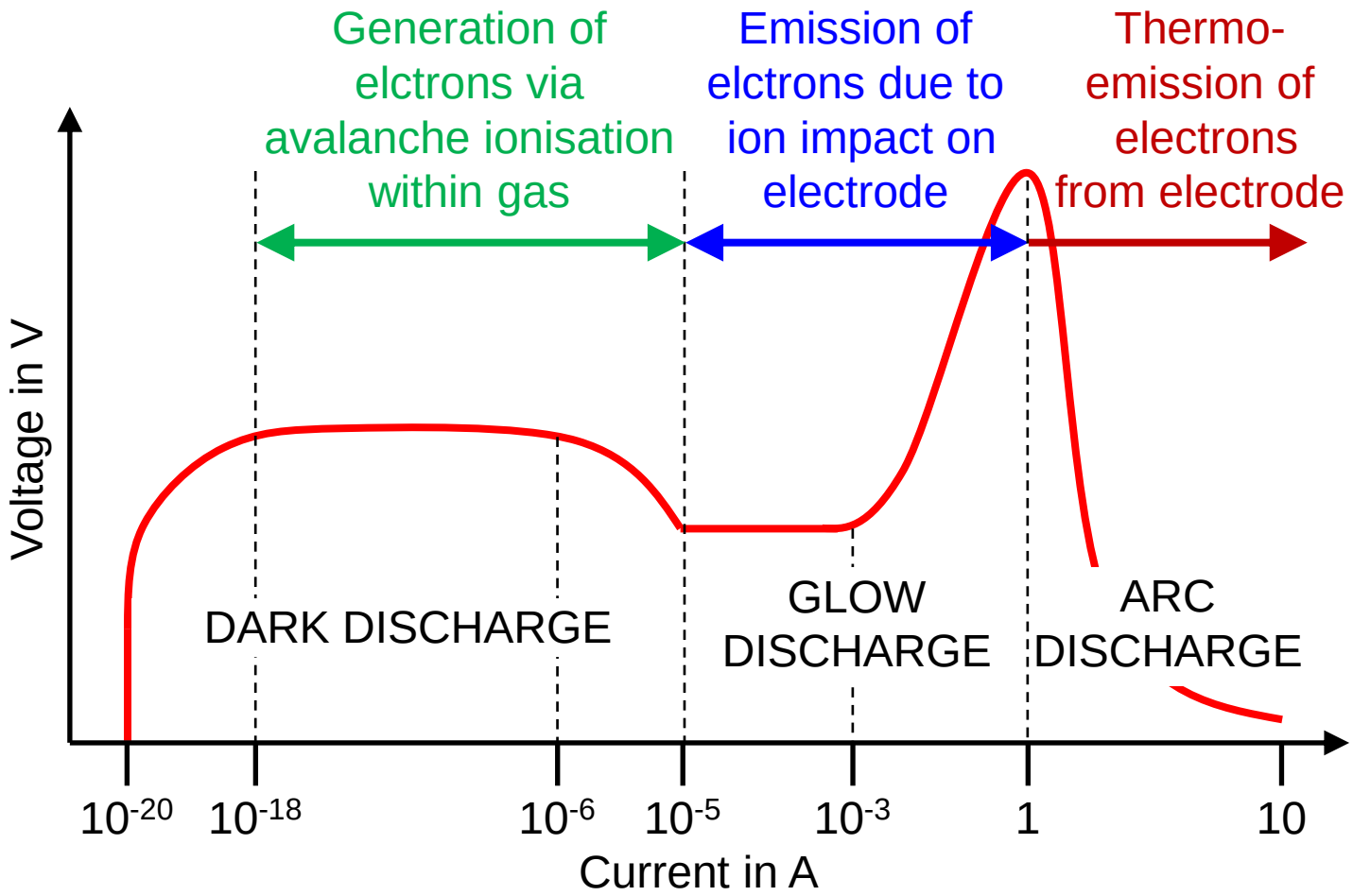
DC discharges

- **Current-voltage-characteristic**
  - …features discrete characteristic zones and inflection points:



DC discharges

- **Current-voltage-characteristic**

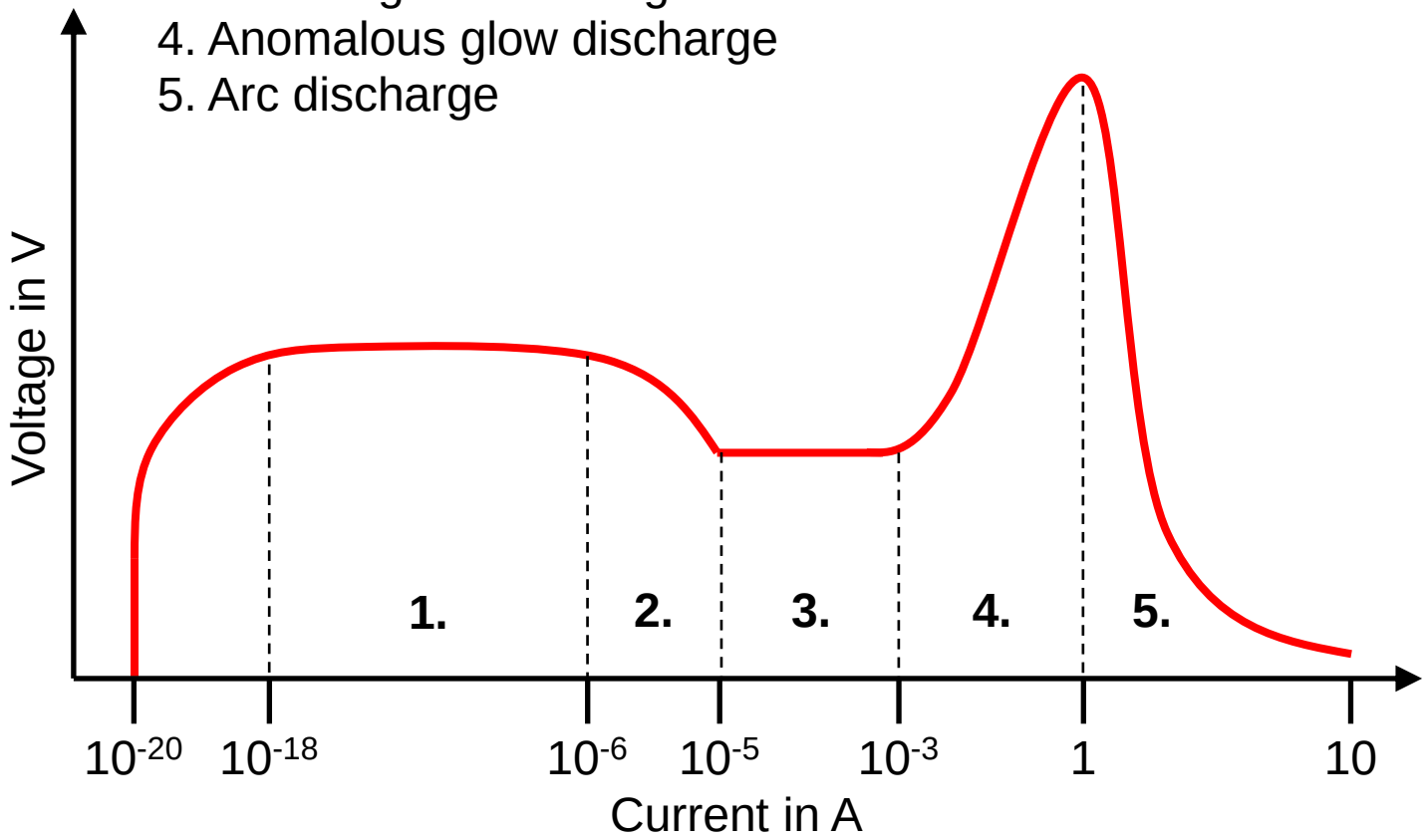


(note true to scale and approximate values!)

DC discharges

- **Current-voltage-characteristic for classification of discharge**

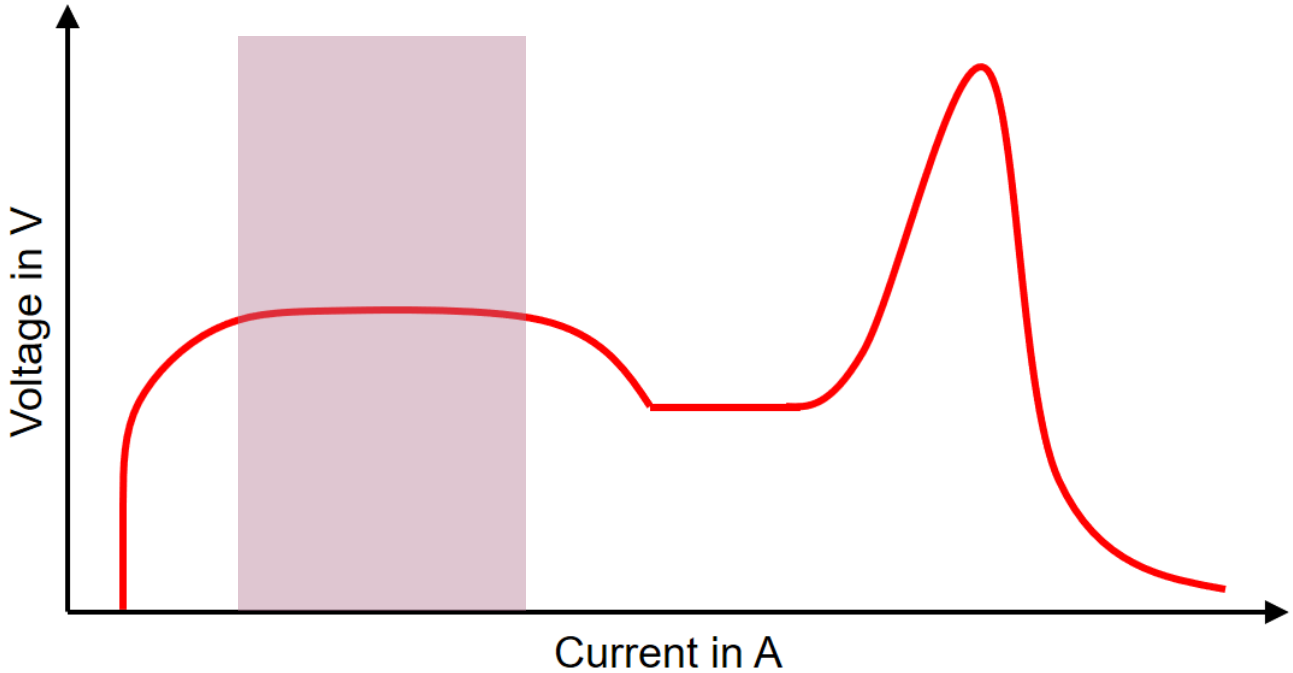
1. Townsend discharge
2. Transition region, aka sub-normal glow discharge
3. Normal glow discharge
4. Anomalous glow discharge
5. Arc discharge



(note true to scale and approximate values!)

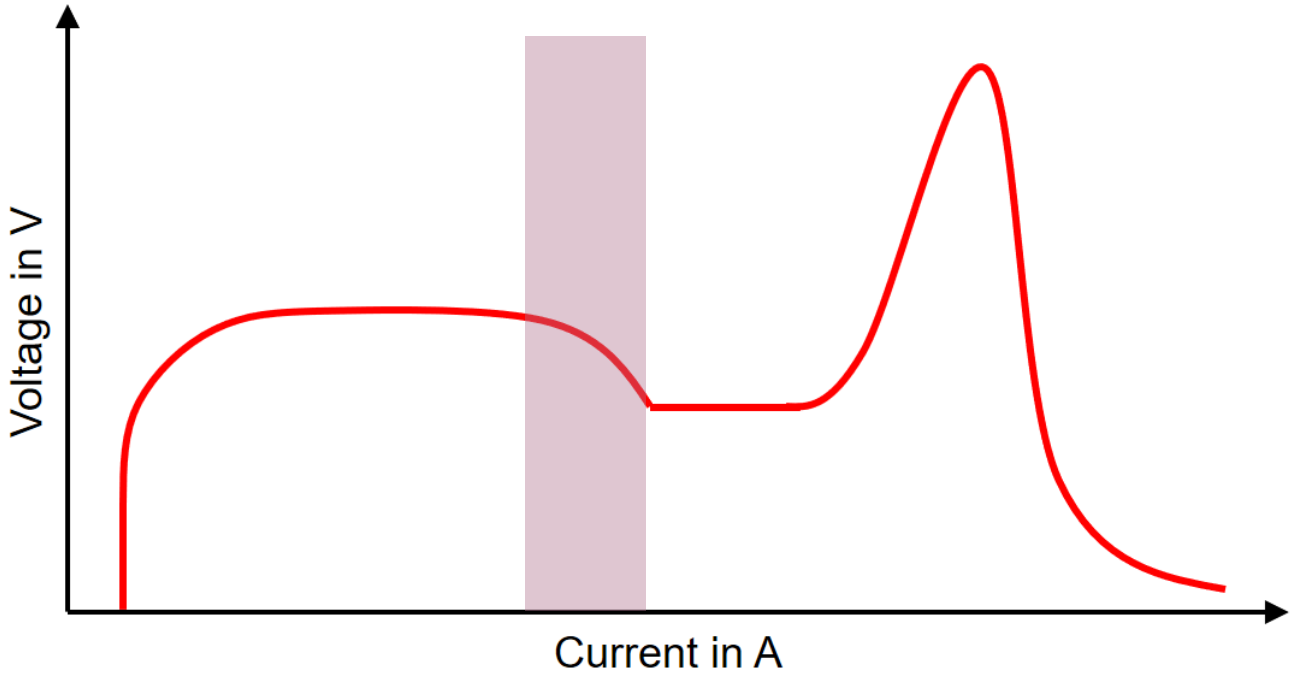
DC discharges

- **Townsend discharge**
  - Non self-maintaining discharge at low currents
  - For higher currents, voltage must be increased in order to increase charge carrier density according to the first Townsend coefficient (see 2.3)



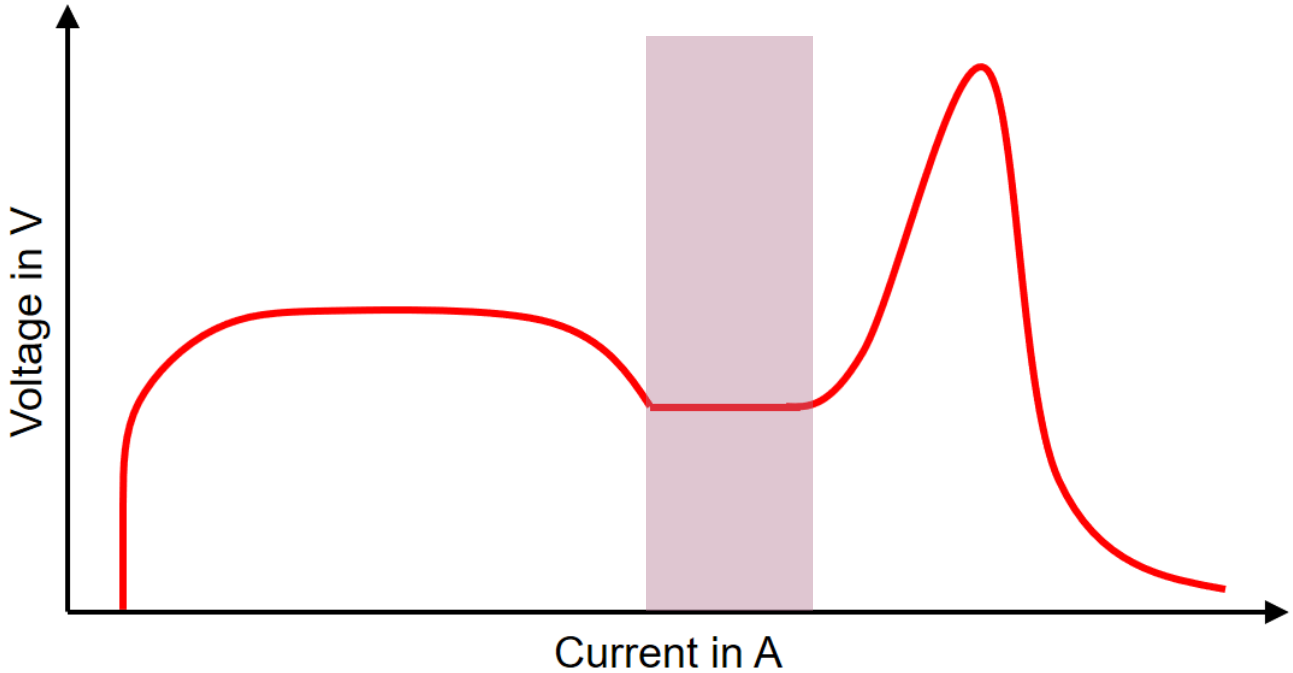
DC discharges

- **Subnormal glow discharge**
  - Voltage is close to, but below breakdown voltage
  - → Small localised non self-maintaining micro discharge, so-called
    - Silent discharge or
    - Corona discharge



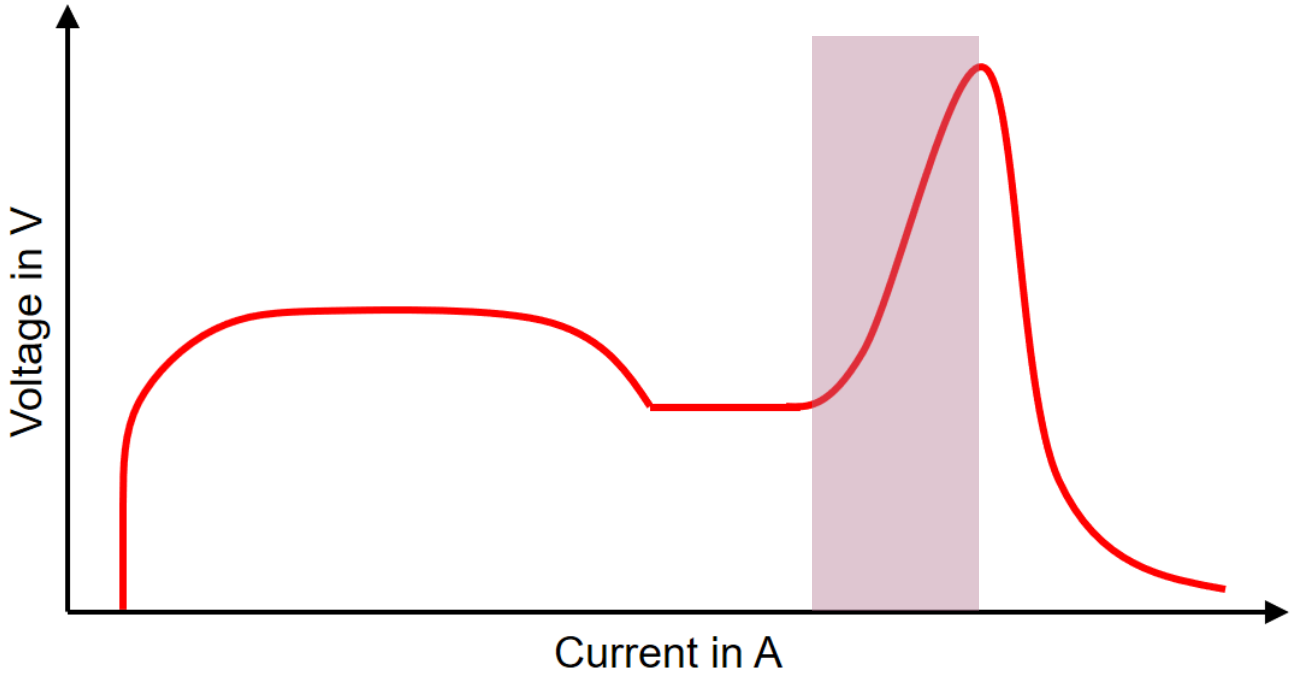
DC discharges

- **Normal glow discharge**
  - Classical DC discharge
  - Criterion for ignition according to Paschen's law (see 2.4.1) is obtained
  - → Self-maintaining discharge
  - Increase in current leads to spatial expansion of plasma



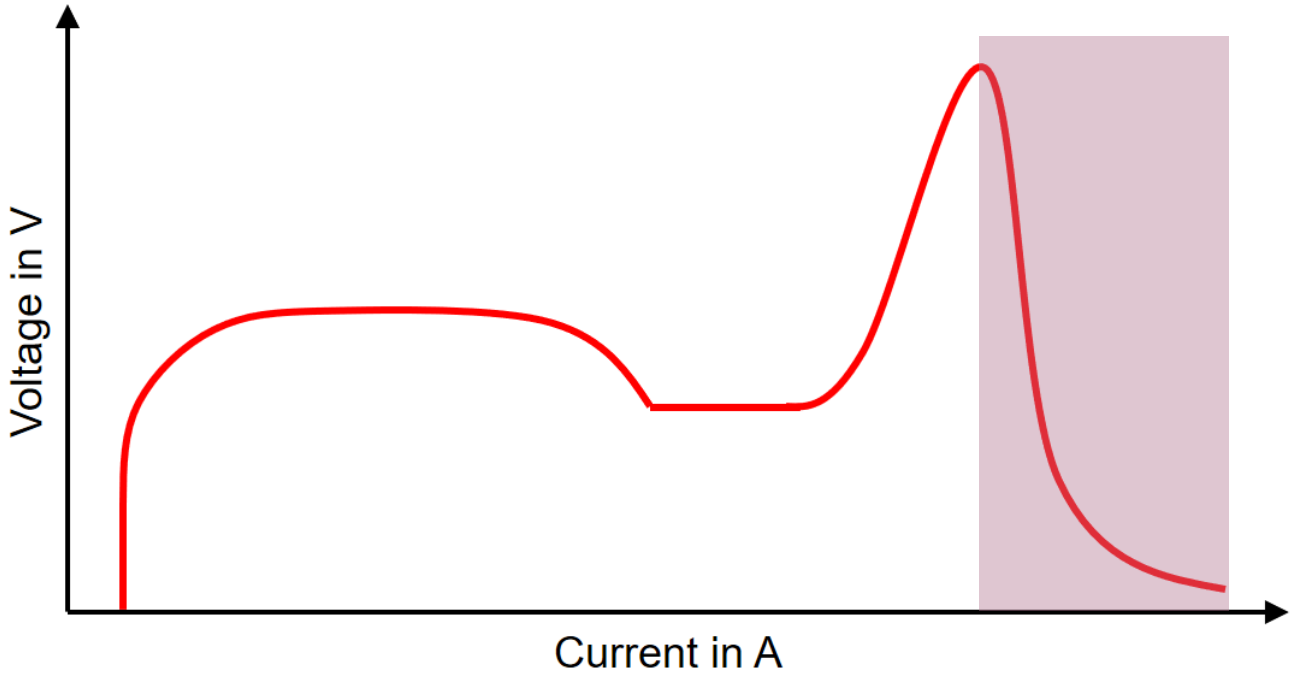
DC discharges

- **Anomalous glow discharge**
  - Spatial expansion of plasma leads to spreading of discharge over areas on the cathode surface with higher distances to the anode
  - Increase in discharge current
  - For entire coverage of electrodes, a further increase in current can only be obtained by increasing voltage
  - Increase in current is proportional to $V^{3/2}$ (Child-Langmuir-law)



DC discharges

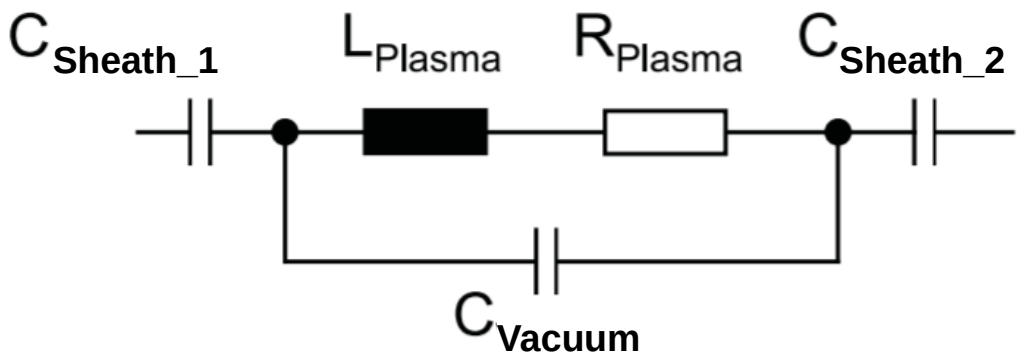
- **Arc discharge**
  - High current leads to notable heating of cathode surface
    → Thermionic emission of further charge carriers into discharge gap
  - Increase in current and accompanying decrease in voltage



DC discharges

???

- **Exercise 2.2**
  - As a simple approach, a plasma source can be described by an equivalent circuit diagram as shown below, schematising a capacitive plasma generator.



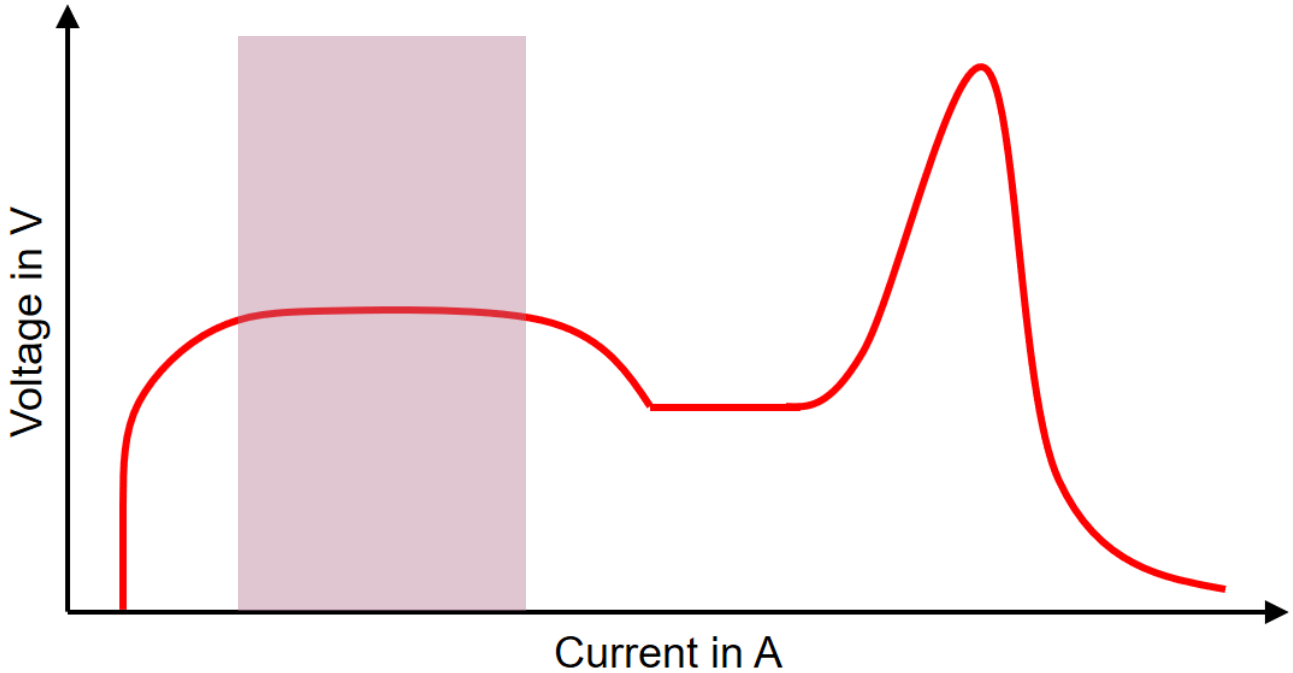
  - We now assume that a plasma source is merely characterised by an electrical resistance of R = 1.7 MΩ. It is moreover operated at an applied voltage of U = 170 V.
    a) Which type of plasma is ignited in this case?
    b) Which type of plasma is ignited if the same voltage is applied, but the resistance is merely R = 90 Ω?

2.3 The Townsend discharge

…where it all begins…

Review: DC discharges

- **Townsend discharge**
  - Non self-maintaining discharge at low currents
  - → Required initial electron for subsequent ionisation is provided by external source (e.g. cosmic irradiation)



The Townsend discharge

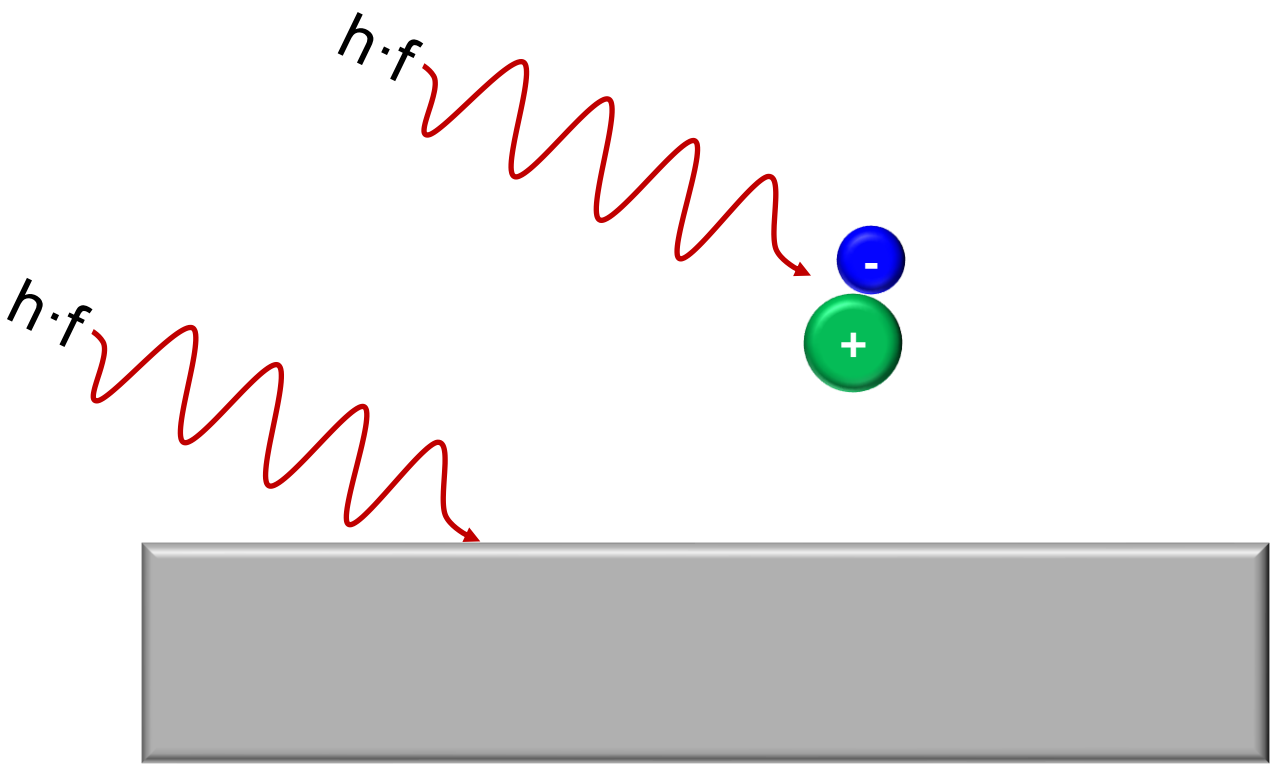
- **One free electron – the ‘booster detonation’ of each plasma discharge**
  - A plasma discharge is initially induced by a single free electron, which starts plasma formation via electron collision-induced ionisation or dissociation.

Where does this initial electron come from?

https://studyflix.de/ingenieurwissenschaften/elektron-2272

The Townsend discharge

- **One free electron – the 'booster detonation' of each plasma discharge**
  - Generation of free electron via
    - Cosmic irradiation (in gas) or
    - Photoelectric emission (from solid, e.g. cathode)



The Townsend discharge

- **One free electron – the 'booster detonation' of each plasma discharge**
  - Condition for ionisation via cosmic irradiation (gas)
    - Energy of incident cosmic irradiation E is higher than or equal to ionisation energy of gas atom or molecule E_{ion}
    - (Note: Atoms and molecules feature different ionisation energies, example hydrogen:
      - Atomic: E_{ion} = 13.6 eV
      - Molecular: E_{ion} = 15.4 eV)

Some helpful constants and interrelations:

$$h = 6.63 \times 10^{-34}\ \text{Js}$$

$$V = \lambda f$$

The speed of light is a constant c, so

$$v = c$$

$$\frac{V}{\lambda} = f$$

$$\frac{c}{\lambda} = f$$

$$c = 3.00 \times 10^{8}\ \text{ms}^{-1}$$

$$h = \frac{E}{f}$$

$$E = hf$$

$$E = h\frac{c}{\lambda}$$

$$\lambda = \frac{hc}{E}$$

The Townsend discharge

???

- **Exercise 2.3**
  - Molecular nitrogen is exposed to cosmic irradiation with a wavelength of 62 nm. Is molecular nitrogen ionised or not by the incident cosmic irradiation?

  - Practice tip: NIST Chemistry WebBook: https://webbook.nist.gov/cgi/cbook.cgi?ID=C7727379&Mask=20

The Townsend discharge

- **One free electron – the 'booster detonation' or 'big bang' of each plasma discharge**
  - Condition for ionisation via photoelectric emission (solid)
    - Energy of incident photon E_{ph} is higher than or equal to work function W of solid material
    - Work function = minimum energy required to remove an electron from an uncharged* solid

Work function of solids commonly used as electrode material in plasma sources/generators:

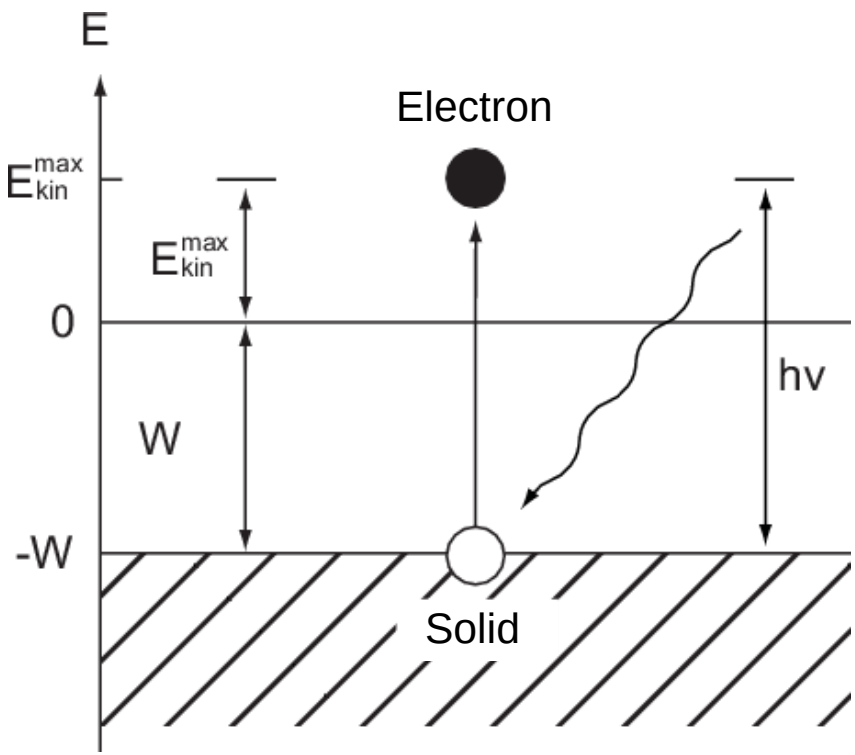
Material	**Work function in eV ($W = f(T)$)**
Copper	4.53 – 5.10
Aluminium	4.06 – 4.26
Iron	4.67 – 4.81

*In electrically charged solids, which applies to the electrode surfaces of a plasma source, the energy required to remove an electron differs from the work function – it is lower according to the Schottky effect. Thus, the work function is the 'worst case' energy required.

The Townsend discharge

- **One free electron – the 'booster detonation' of each plasma discharge**
  - Condition for ionisation via photoelectric emission (solid)
    - (Kinetic) energy of removed electron E_{e-} or E_{kin}, respectively, is given by

$$E_{e-} = E_{kin} = E_{ph} - W$$



The Townsend discharge

???

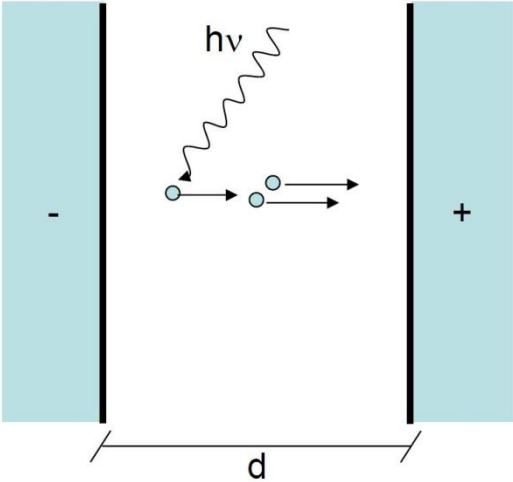
- **Exercise 2.4**
  - Determine the energy of an electron, which is generated via photoelectric emission at a copper surface where the wavelength of incident light is 210 nm.

The Townsend discharge

???

- **Exercise 2.5**
  - Determine the wavelength of incident light required to remove an electron from an aluminium surface via photoelectric emission where the resulting energy of the removed electron should be sufficient for ionising atomic hydrogen.

The Townsend discharge



- Application of voltage at gas-filled gap

→ Charge carriers can be accelerated and induce ionisation of the gas

- If the very first charge carrier is generated by an external source (e.g. cosmic radiation or photoelectric emission)

→ **non-self-maintained discharge**

- Electron induces ionisation process within gas according to avalanche model (preferentially at low gas pressure *p* and product of pressure and distance *pd*, respectively)

→ **Townsend mechanism**

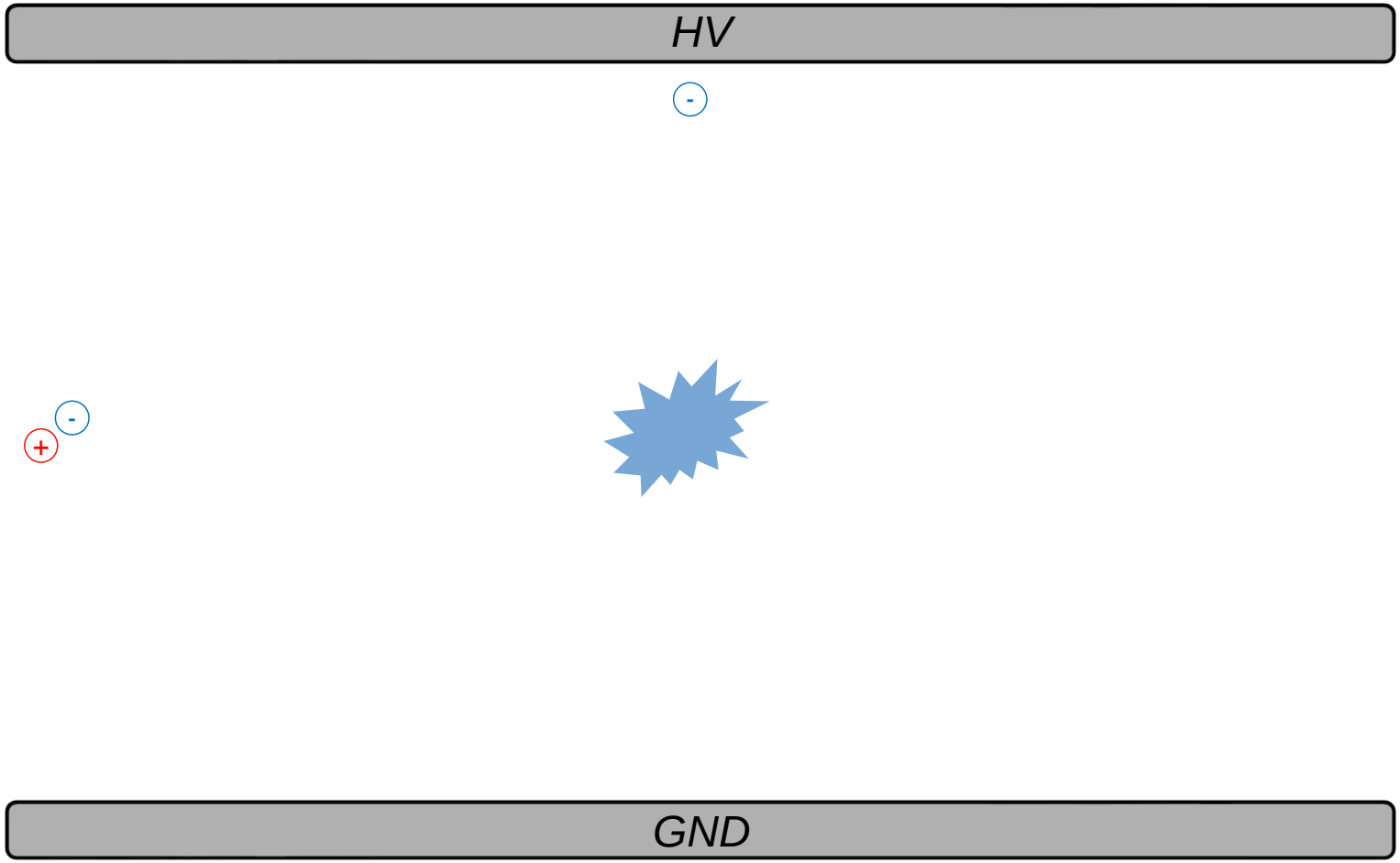
- Nearly no excitation and corresponding photo emission due to comparatively low electron density
- Applied voltage lower than ignition voltage → low electrical current
- Ionisation of gas if electron reaches sufficient energy for ionisation within available (gas-dependent) free length of path

→ Each new electron causes further ionisation an thus generation of free electron

→ Formation of **electron avalanche**

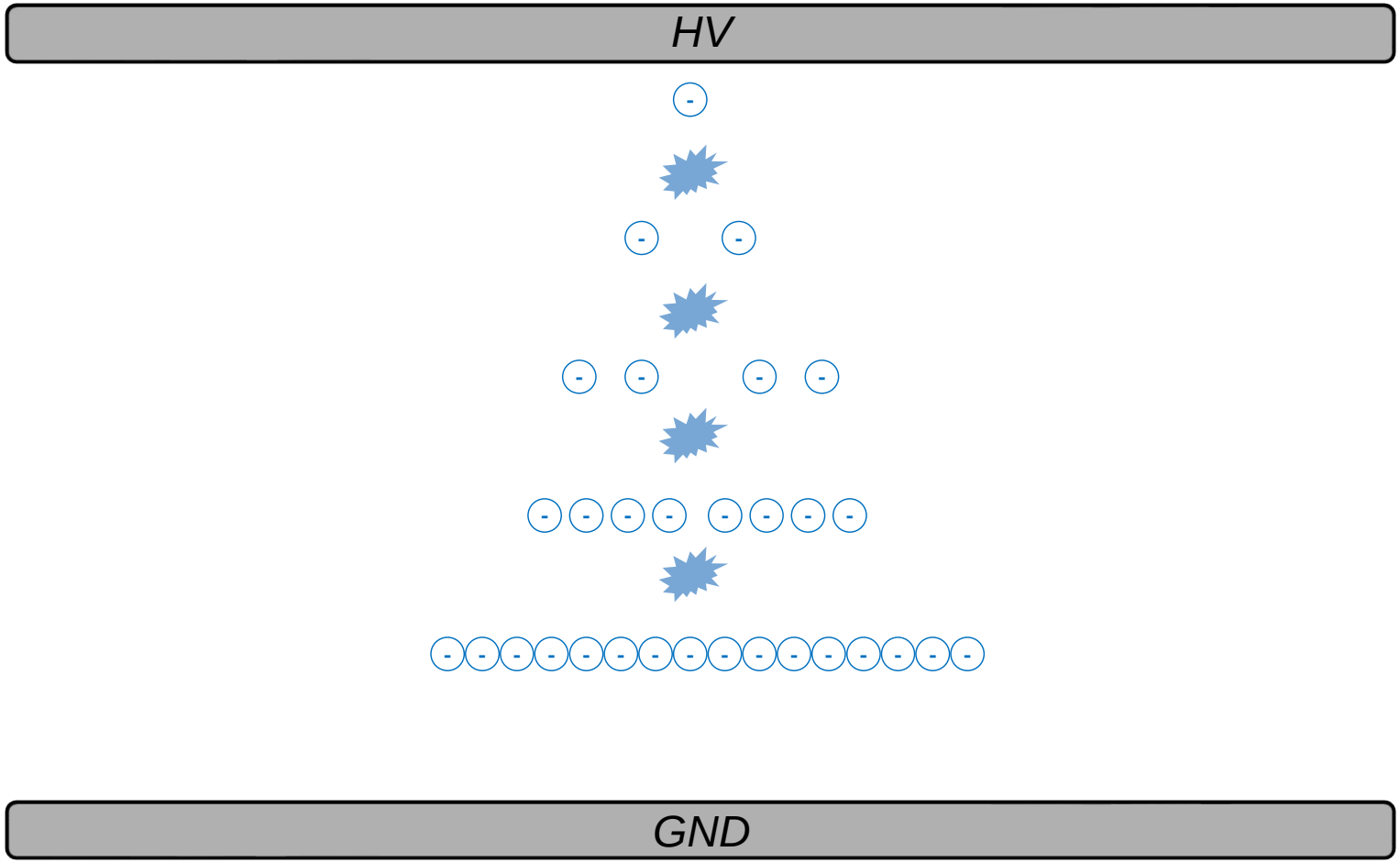
The Townsend discharge

- **Electron avalanche**
  - Electrical field accelerates electron
    → Generation of further free electrons via collision and ionisation

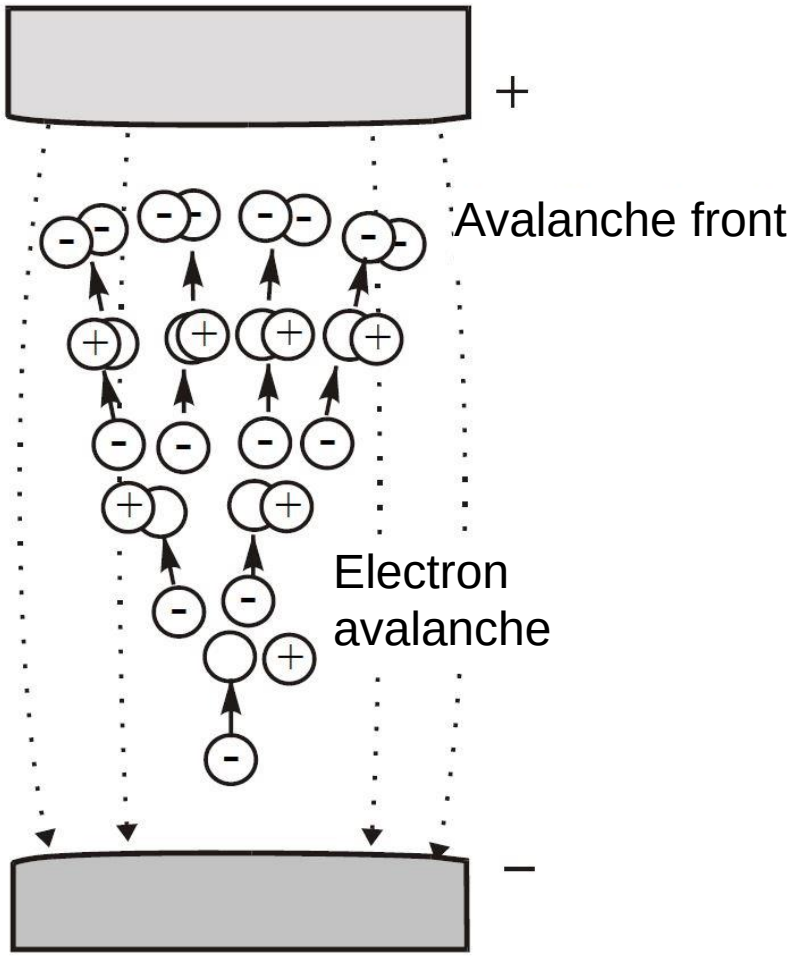


The Townsend discharge

- **Electron avalanche**
  - Successive generation of free electrons → electron avalanche



The Townsend discharge



- Electron avalanche propagates towards anode

→ Inelastic scattering of electron at each collision in case of sufficient electron energy

→ Increase in current per distance dx at an amplification factor α und and an external source term S_{extern}:

$$\mathrm{d}\Gamma = \alpha\Gamma\mathrm{d}x + S_{extern}\mathrm{d}x$$

→ Electron flux:

$$\Gamma = S_{extern}\frac{1}{\alpha}(\exp\alpha x - 1)$$

α is the **<u>1rst Townsend coefficient</u>**

Note: The electron energy E_e-, which can be collected by an electron between two collisions, follows from the free length of path and the field strength according to

$$E_{e^-} = \lambda \cdot \vec{E}$$

The Townsend discharge

- 1rst Townsend coefficient = number of ionisation incidents per distance and electron, given by

$$\alpha = Ap \exp\left(-B\frac{p}{E}\right)$$

→ The multiplication of electrons and thus the first Townsend coefficient *α* are a function of the electric field strength *E* and pressure *p* – and particle density *n*, respectively.

- For sufficient discharge gap thickness *d* and voltage *V*, where E=V/d:

$$\alpha = Ap \exp\left(-B\frac{pd}{V}\right)$$

The Townsend discharge

$$\alpha = Ap \exp\left(-B\frac{pd}{V}\right)$$

- The first Townsend coefficient depends on
  - Electric field strength and voltage, respectively,
  - Pressure AND
  - Type of gas and/or gas mixture

- The gas-dependent coefficients A and B can be determined on the basis of the ion radius r_I, the ionisation energy E_I, the Boltzmann constant k_B ,and the elementary charge e:

$$A = \frac{\pi r_I^2}{k_B T}$$

$$B = \frac{\pi r_I^2 E_I}{T k_B e}$$

Gas (mixture)	**A in (Pa·m)$^{-1}$**	**B in V/(Pa·m)**
Helium	2.25	25.50
Hydrogen	3.83	104.10
Nitrogen	9.00	256.5
Argon	10.20	176.30
Air	10.95	273.8
Carbon dioxide	15.00	349.50

Jane Lehr, Pralhad Ron: Electrical breakdown in Gases. In: Foundations of pulsed power technology. John Wiley & Sons, Inc., 2017

The Townsend discharge

- **The Townsend coefficients**
  - Named after John Sealy Townsend
  - Description of properties of gas discharges under influence of an external electrical field in plasma physics

  - **1^{rst} Townsend coefficient α:**
    - Probability per length of path, where new free electrons are generated/removed from an atom or molecule via collisions with other charge carriers, i.e., electrons or ions

  - Townsend coefficient β:
    - Probability per length of path, where new free electrons are generated via collisions between neutrals, i.e., atoms or molecules (usually negligible, since notably small with respect to α!)

  - **2nd Townsend coefficient γ:**
    - Material constant of the electrode material, specifying the number of electrons released by the electrode material per incident ion (see 2.4)

The Townsend discharge

- **Noteworthy**
  - The reduced electric field strength, i.e., the ratio of the electric field strength and the particle density (E/n) is usually given in the unit Townsend Td = 10^{-17} Vcm^2.

- **Summary**
  - The first electron generates an electron avalanche
  - Free electrons lead to the generation of photons, ions and metastable neutrals (via electron collision), that can generate further free electrons when hitting the electrode surfaces (via photoelectric emission, sputtering, and de-excitation and accompanying release of energy*)

The Townsend-mechanism represents the initial stage of plasma discharges!

*For instance, argon metastables carry an energy of about 11 eV, which is released via collisions at walls.

The Townsend discharge

???

- **Exercise 2.6**
  - In a plasma source, the discharge gap, i.e., the distance between the electrodes, amounts to 1 cm. We assume that an initial electron, which is emitted by the cathode, causes ionisation, and that each electron resulting from such ionisation features sufficient energy for ionisation of the gas (here: a mixture of 80 % nitrogen and 20% oxygen) between the electrodes. The plasma is operated at
    a) atmospheric pressure,
    b) a pressure of 1 hPa, and
    c) a pressure of 10^{-7} hPa.

  - Estimate the minimum number of free electrons generated between both electrodes in case a), b), and c).

The Townsend discharge

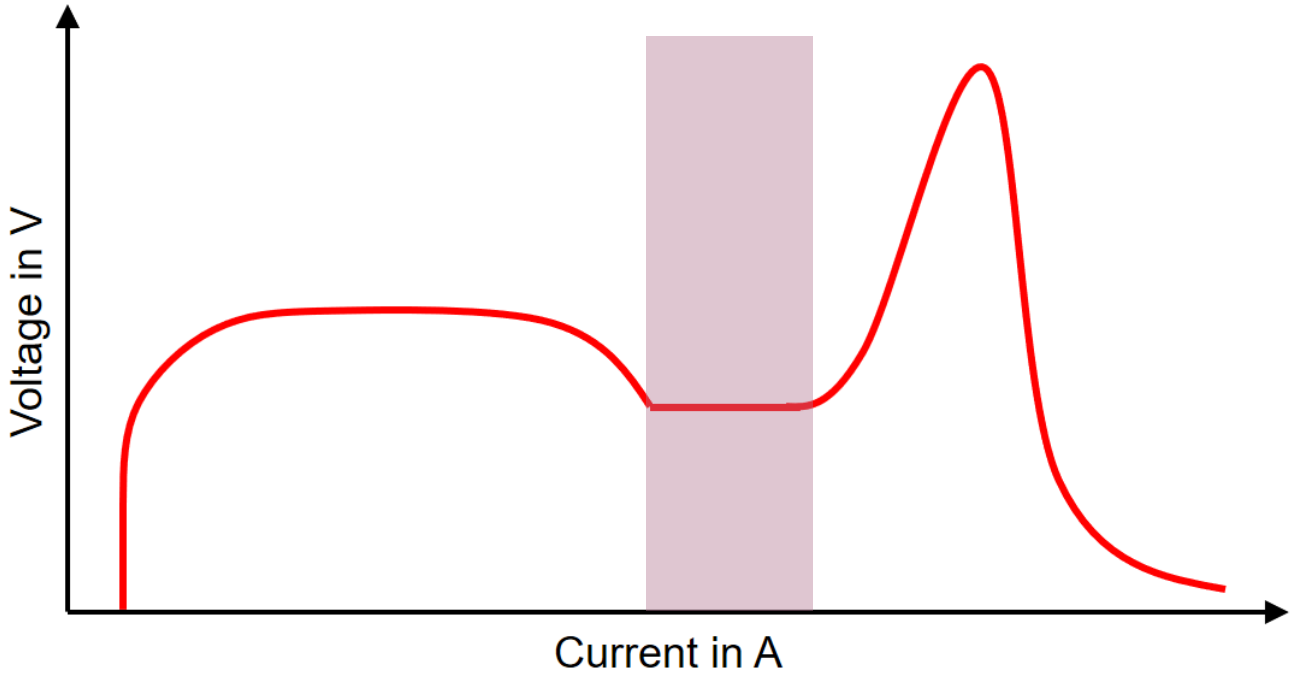
???

- **Exercise 2.7**
  - A plasma is ignited in pure molecular hydrogen as process gas. The discharge gap width is 2.54 cm and the applied voltage amounts to 12 kV. Determine the 1^{rst} Townsend coefficient for this configuration.

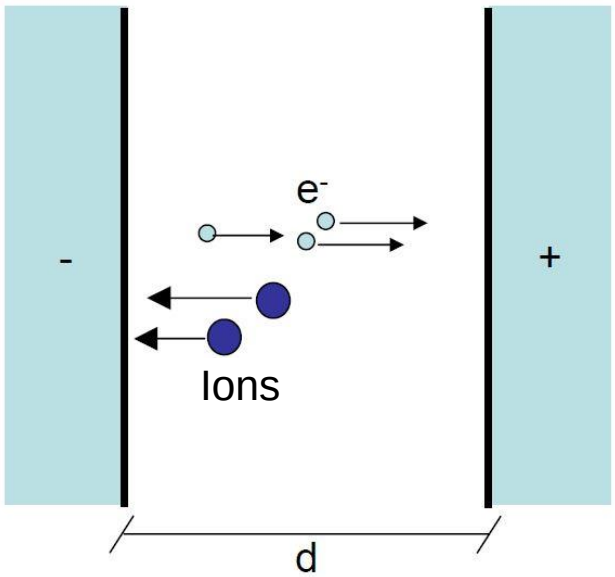
2.4 The glow discharge

Review: DC discharges

- **(Normal) glow discharge**
  - Criterion for ignition according to Paschen's law (see 2.4.1) is obtained
  - → Self-maintaining discharge



The glow discharge



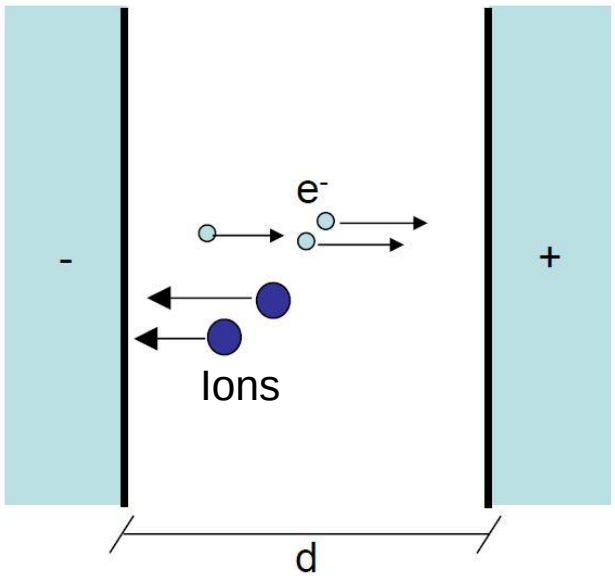
- Townsend discharges are non-self-maintained discharges.
- For the ignition of a **<u>self-maintained discharge</u>** within a gas-filled gap, two conditions should generally be fulfilled
    1. Electrons must have collected sufficient energy for inducing ionisation
    2. Multiplication of electrons is characterised by coefficient α, the local electron flux is thus:

$$\frac{\mathrm{d}\Gamma_e}{\mathrm{d}x} = \alpha\Gamma_e \quad \text{and} \quad \Gamma_e(d) = \Gamma_e(0)e^{\alpha d}.$$

→ Number of generated electrons within this avalanche:

$$\Gamma_e(d) - \Gamma_e(0) = \Gamma_e(0)e^{\alpha d} - \Gamma_e(0)$$

The glow discharge

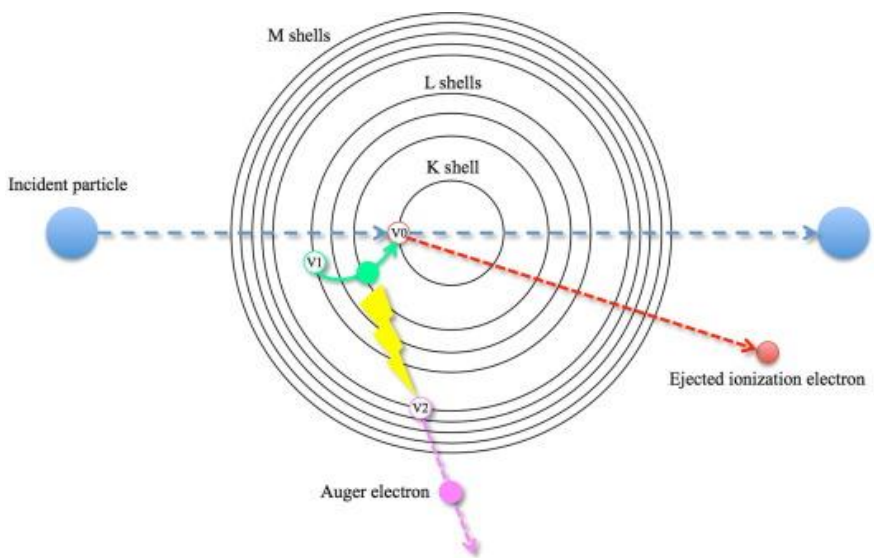


- Each ionisation event does not only generate a free electron, but also an ion:

$$\Gamma_i(0) - \Gamma_i(d) = \Gamma_e(d) - \Gamma_e(0)$$

→ A discharge becomes self-maintained if the generation of the first electrons according to $\Gamma(0)$ does not depend on an external source such as cosmic radiation

- Possible processes:
  - Generation of secondary electrons via impact of ions on cathode (= sputtering)
  - Release of Auger electrons
  - Release of electrons due to neutralisation of ions at solid surfaces (see next slide…)



The glow discharge

- Ions hitting a dielectric solid at a low kinetic energy are neutralised by electrons from the valence band of the solid
- The energy released during neutralisation can be transferred to a second electron within the solid, which may then be emitted by the cathode surface
- Ion current and generation of secondary electrons are interrelated via

$$\Gamma_e(0) = \gamma_i \Gamma_i(0)$$

- Secondary electrons emitted by the anode cannot be accelerated ('wrong electrode') and do thus not contribute to the plasma ignition process
- Assuming $\Gamma_i(d) = 0$ leads to

$$\frac{1}{\gamma_i}\Gamma_e(0) = \Gamma_e(0)e^{\alpha d} - \Gamma_e(0)$$

- γ is the **<u>2nd Townsend coefficient</u>** → = material constant of the electrode material, specifying the number of electrons released by the electrode material per incident ion
- Condition for ignition: $\alpha d = \ln\left(1 + \frac{1}{\gamma}\right)$

The glow discharge

- Coming from the first Townsend coefficient, which contains the multiplication of electrons (= *f*(*E*, *p*, *n*) it can be written:

$$\alpha \propto n_g \exp\left(-\frac{E_{Ionisation}}{E\lambda}\right) = n_g \exp\left(-\frac{E_{Ionisation}}{V/d\lambda}\right)$$

- V/d: describes energy, which can be collected by an electron between two collisions
- The given conditions

$$n_g \propto p \text{ and } \lambda \propto p^{-1} \text{ and}$$

$$\alpha = Ap \exp\left(-B\frac{pd}{V}\right)$$

leads to:

$$Apd \exp\left(-B\frac{pd}{V}\right) = \ln\left(1 + \frac{1}{\gamma}\right)$$

2.4 The glow discharge

2.4.1 Paschen’s law

Paschen's law

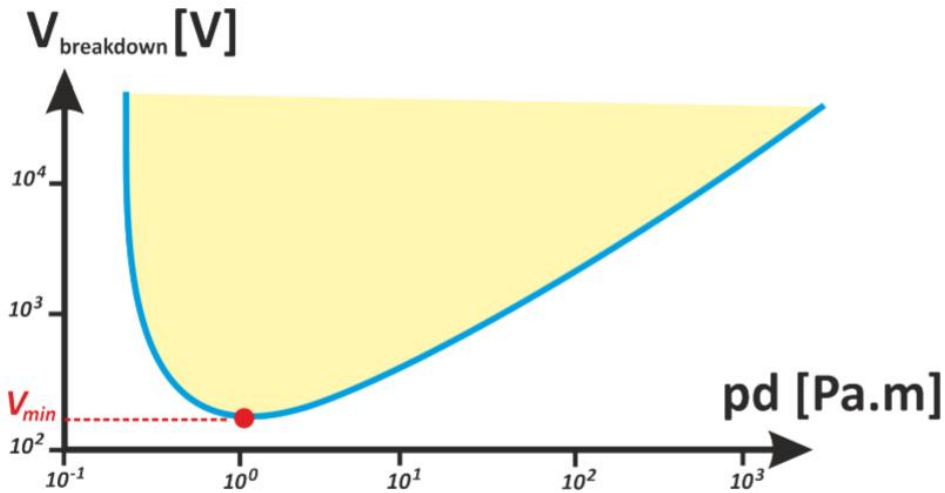
- Solving

$$Apd \exp\left(-B\frac{pd}{V}\right) = \ln\left(1+\frac{1}{\gamma}\right)$$

for voltage U and V, respectively, gives Paschen's law:

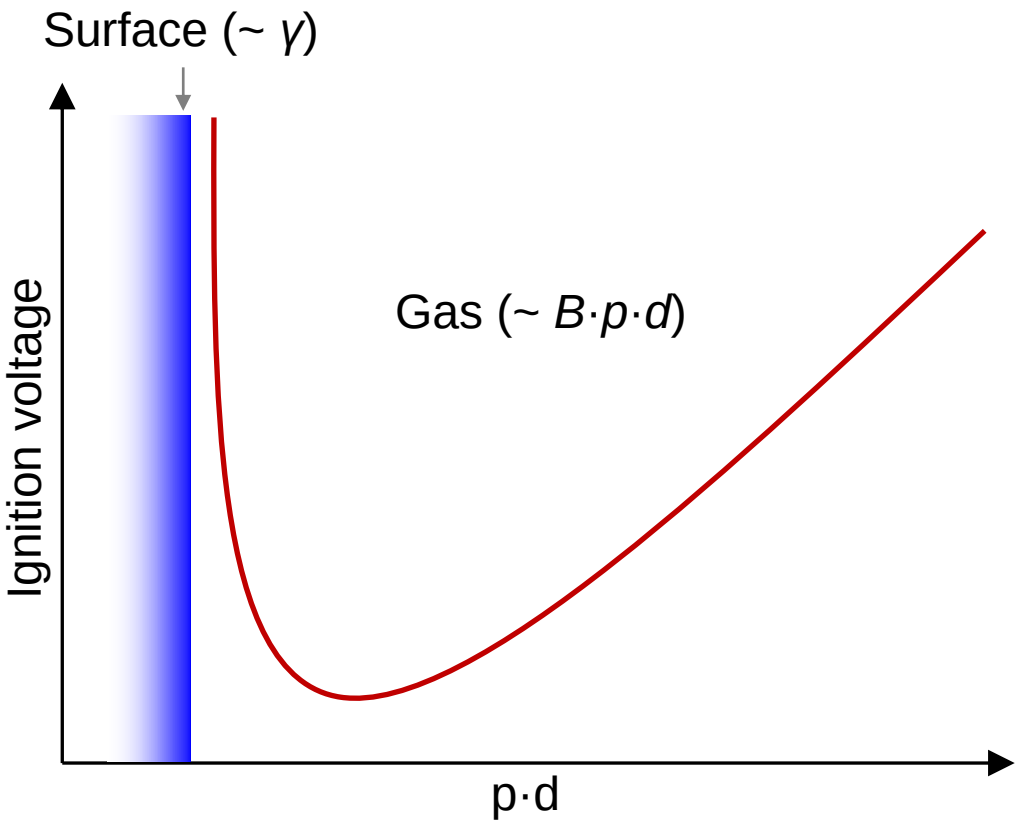
$$U = V = \frac{Bpd}{\ln(Apd) - \ln[\ln(1+\gamma^{-1})]}$$

→ **<u>Paschen's curve</u>**



Paschen's law

$$U = V = \frac{Bpd}{\ln(Apd) - \ln[\ln(1+\gamma^{-1})]}$$



(or d for p = const. and vice versa)

- For high values p·d almost linear interrelationship of ignition voltage and product B·d·p
- B contains ionisation energy (and thus depends on gas)
- Minimum at low values p·d, given by

$$Apd - \ln(1+\gamma^{-1}) = 0$$

and thus dependent on electrode material

→ **Paschen's minimum**

Paschen's law

- *Examples for Paschen's minima*:

Gas	Cathode material	Ignition voltage @ $p \cdot d_{min}$ in V	$p \cdot d_{min}$ in Pa·m
Argon	Iron	265	2.0
Helium	Iron	150	3.3
Mercury	Iron	520	2.6
Mercury	Tungsten	425	2.36
Nitrogen	Iron	275	1.0
Oxygen	Iron	450	0.92

- In a glow discharge, the ignition voltage at the Paschen's minimum corresponds to the cathode fall voltage U_{cf}:

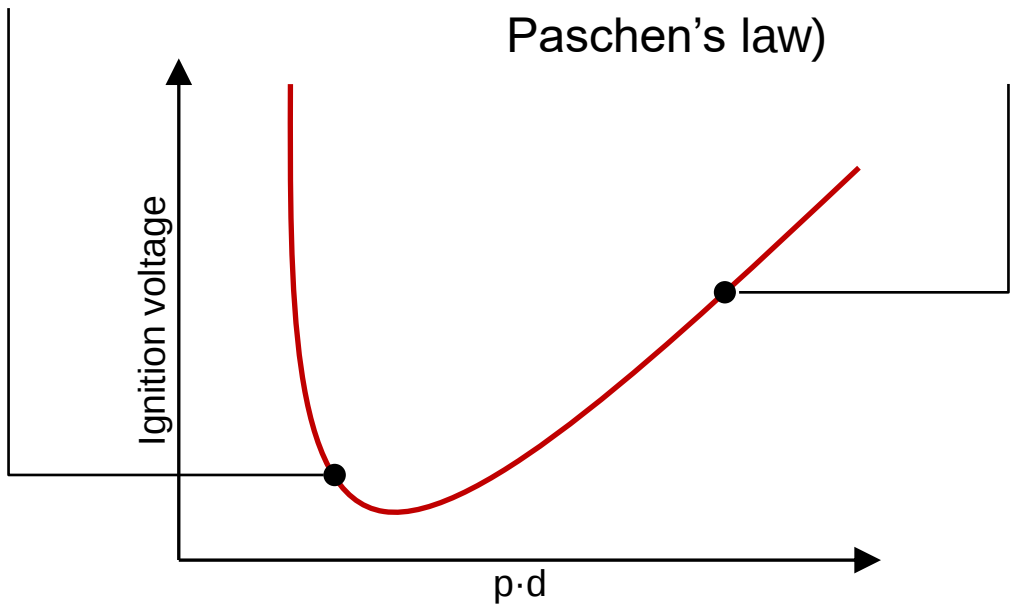
https://commons.wikimedia.org/w/index.php?curid=29160922

Paschen's law

- **Plasma discharge types along Paschen's curve**

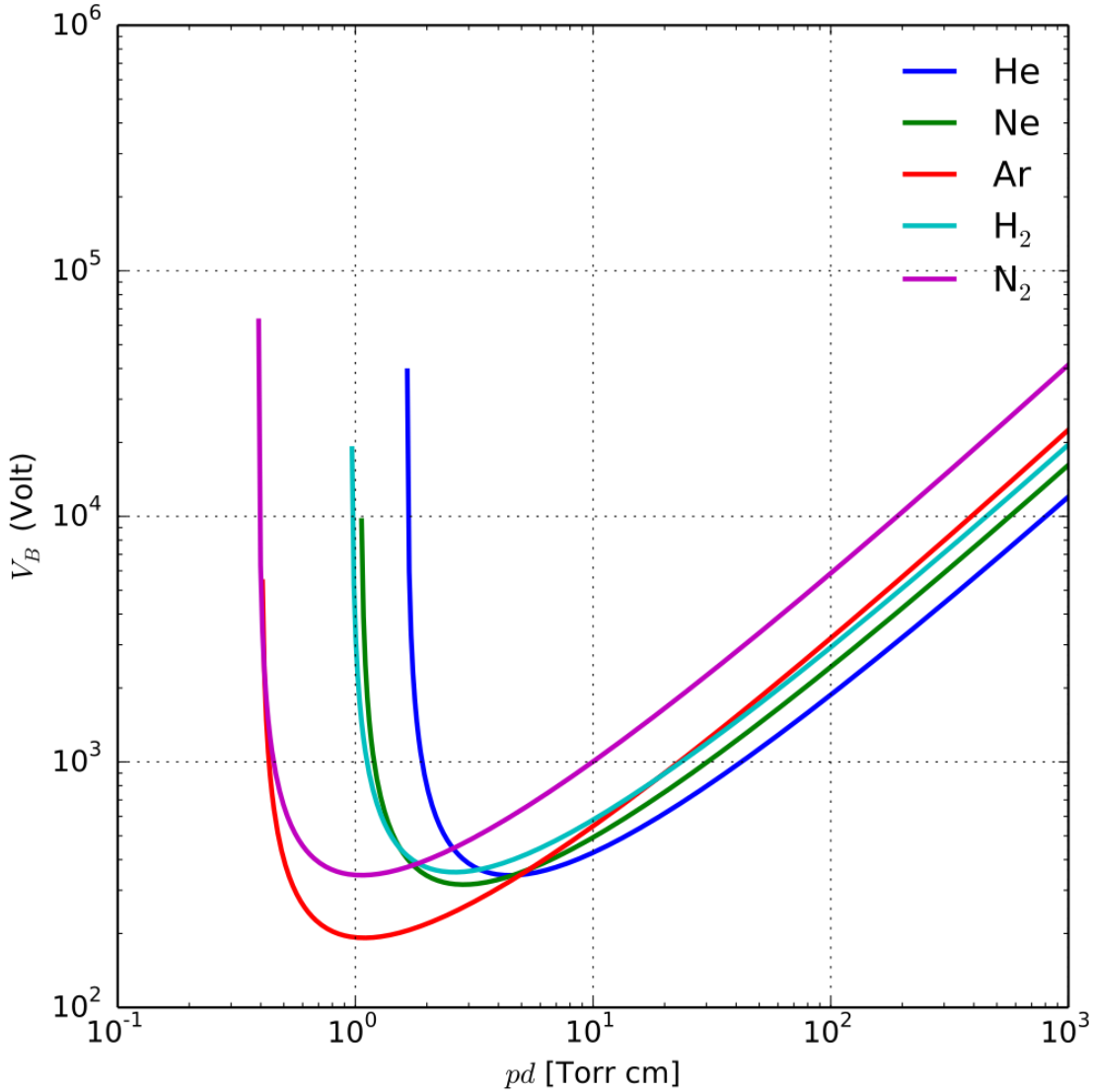
Typical low pressure plasmas for sputtering and etching are found left to Paschen's minimum → discharge does not occur at shortest distance between electrodes due to comparatively high ignition voltage required for short distances

Atmospheric pressure plasmas and high pressure plasmas are found at the linear part of Paschen's curve right to the Paschen's minimum→ discharge occurs at elevated points (e.g. humps) on electrode surface, where p·d and thus required voltage is minimal (in case of high surface roughness, high electric fields occur, restricting the validity of Paschen's law)



Paschen's law

- *Examples for Paschen's curves*:



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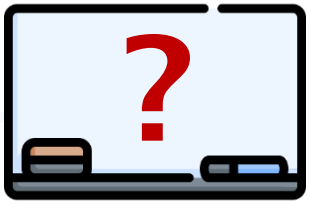
Paschen's law

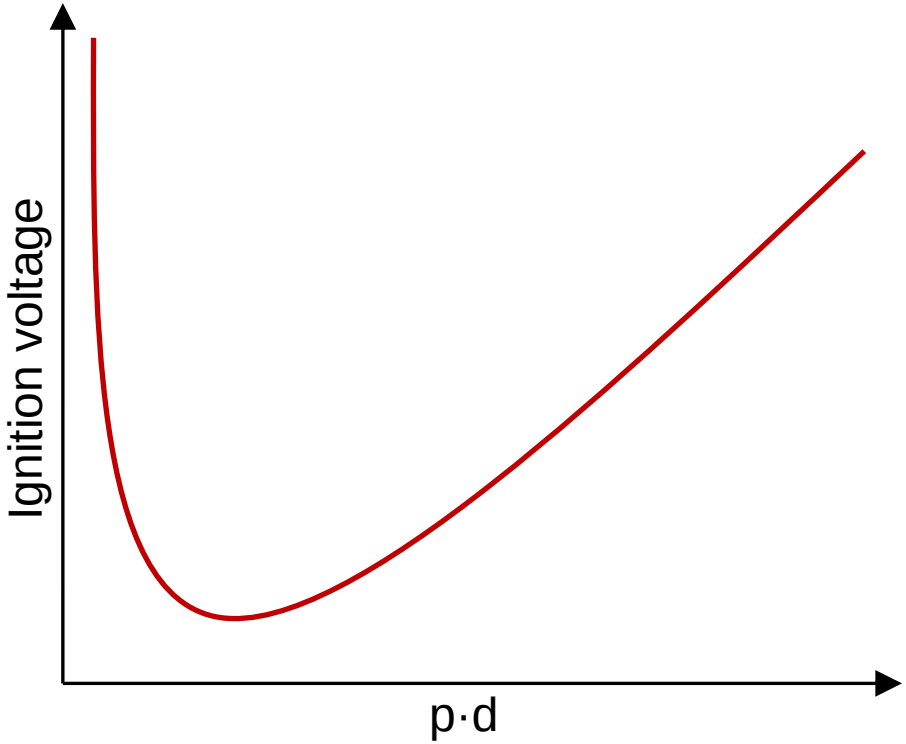


- **Worked example 2.1**
  - In the following table, different values for the ignition voltage and $p \cdot d_{min}$ are displayed for the gas mercury…why?

Gas	**Cathode material**	**Ignition voltage @ $p \cdot d_{min}$ in V**	**$p \cdot d_{min}$ in Pa·m**
Argon	Iron	265	2.0
Helium	Iron	150	3.3
Mercury	Iron	**520**	**2.6**
Mercury	Tungsten	**425**	**2.36**
Nitrogen	Iron	275	1.0
Oxygen	Iron	450	0.92

Paschen's law

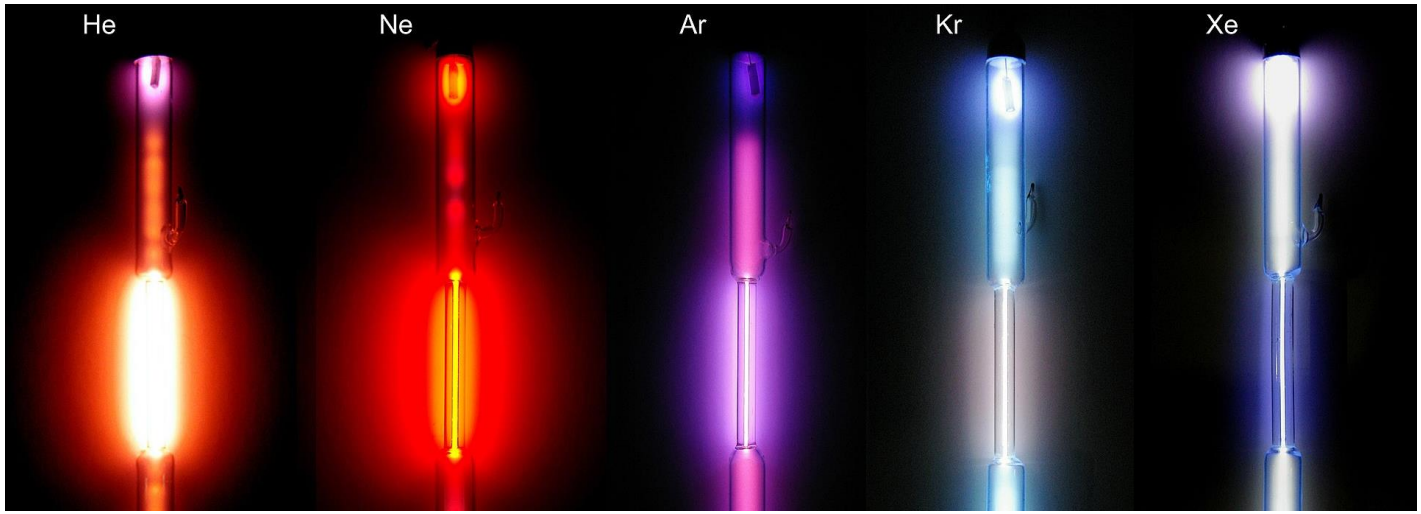
- **Worked example 2.2**
  - As expressed by the Paschen's curve, a very high ignition voltage is required for very low and very high p·d-values (and discharge gap thicknesses in case of constant pressure, respectively)…why?



Paschen's law

- **Worked example 2.3**
  - A light source, based on the principle of fluorescent tubes, shall be realised considering the following requirements:
    - Gas pressure within the tube: 1000 Pa
    - Colour of light emitted by gas discharge: bright red-orange
    - Cathode material: iron
    - Required voltage: as low as possible

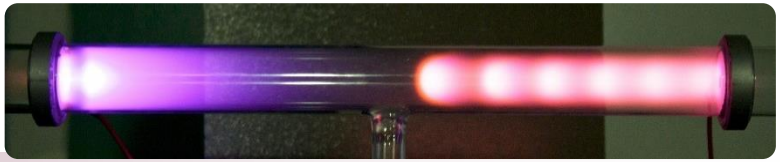
a) Which gas could be used to fill the tube?
b) Determine the optimum length of the tube.



2.4 The glow discharge

2.4.2 Characteristic plasma zones in glow discharges

The glow discharge



- **Properties**
  - Glow discharges
    - are driven at low gas pressure.
    - are operated at higher current than Townsend discharges

  - Neutral gas remains cold.
  - Merely electrons and ions gain high kinetic energy in some particular zones within the discharge
    → Formation of a certain arrangement of space-charge regions, which have a particular function for the discharge
    → Formation of a characteristic structure consisting of several layers or zones with different space-charges

  - Structure allows higher currents in glow discharges in comparison to Townsend discharges

  - Glow discharges feature lower operating voltage than Townsend discharges ignited at the same p·d value

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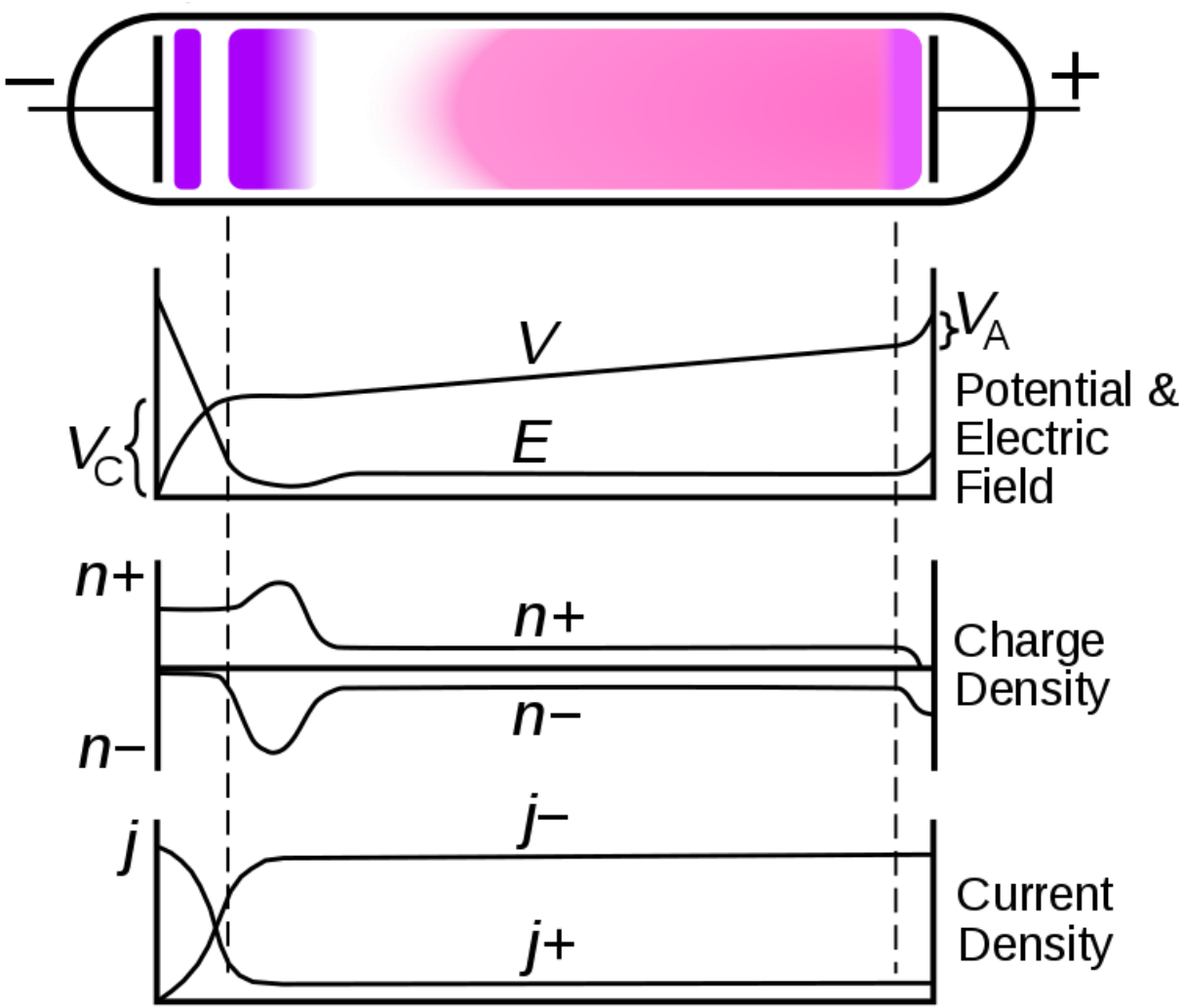
The glow discharge

- **Characteristic zones of a typical glow discharge**



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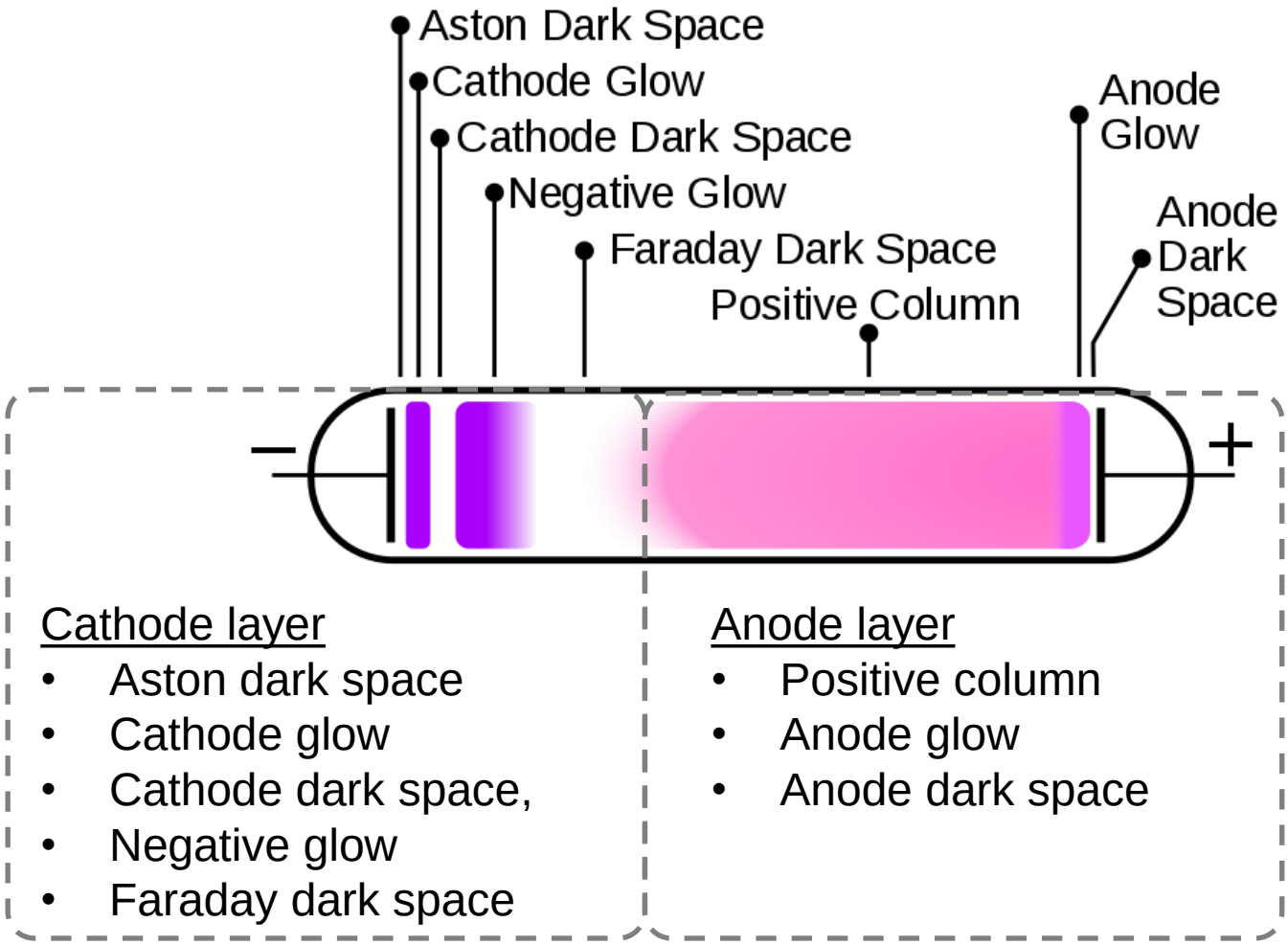
The glow discharge

- **Characteristic zones of a typical glow discharge**



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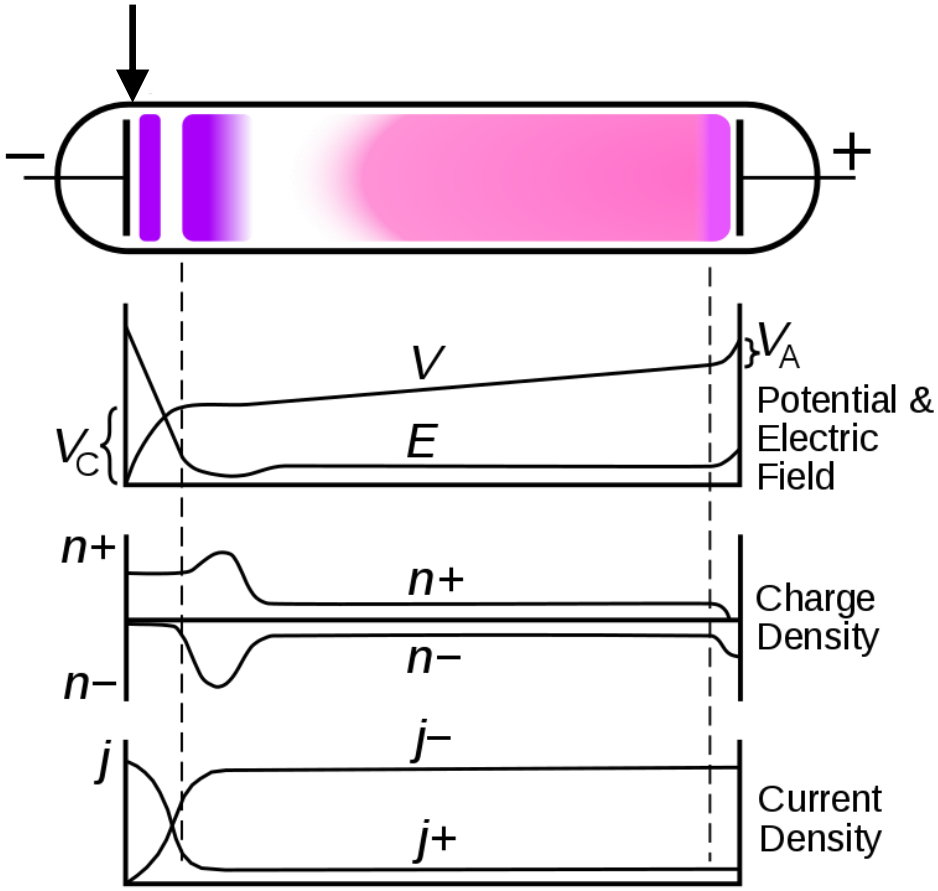
The glow discharge

- **Characteristic zones of a typical glow discharge – the cathode layer**
  - <u>Aston dark space</u>
  - Thickness of layer to small, electrons do not gain sufficient energy for excitation of atoms
    → no de-exciatation and accompanying emission of light



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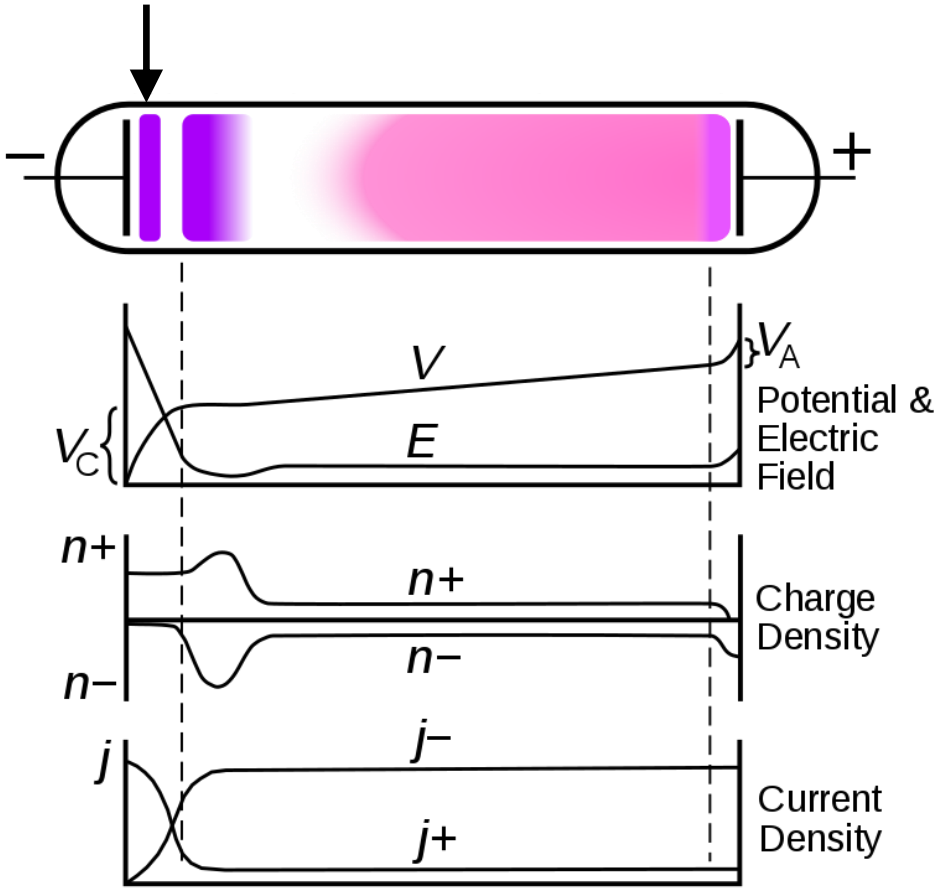
The glow discharge

- **Characteristic zones of a typical glow discharge – the cathode layer**
  - <u>Cathode glow</u>
  - Electrons from cathode eventually feature sufficient energy for excitation of atoms, leading to emission of light with gas-dependent colour via spontaneous de-excitation very close to cathode



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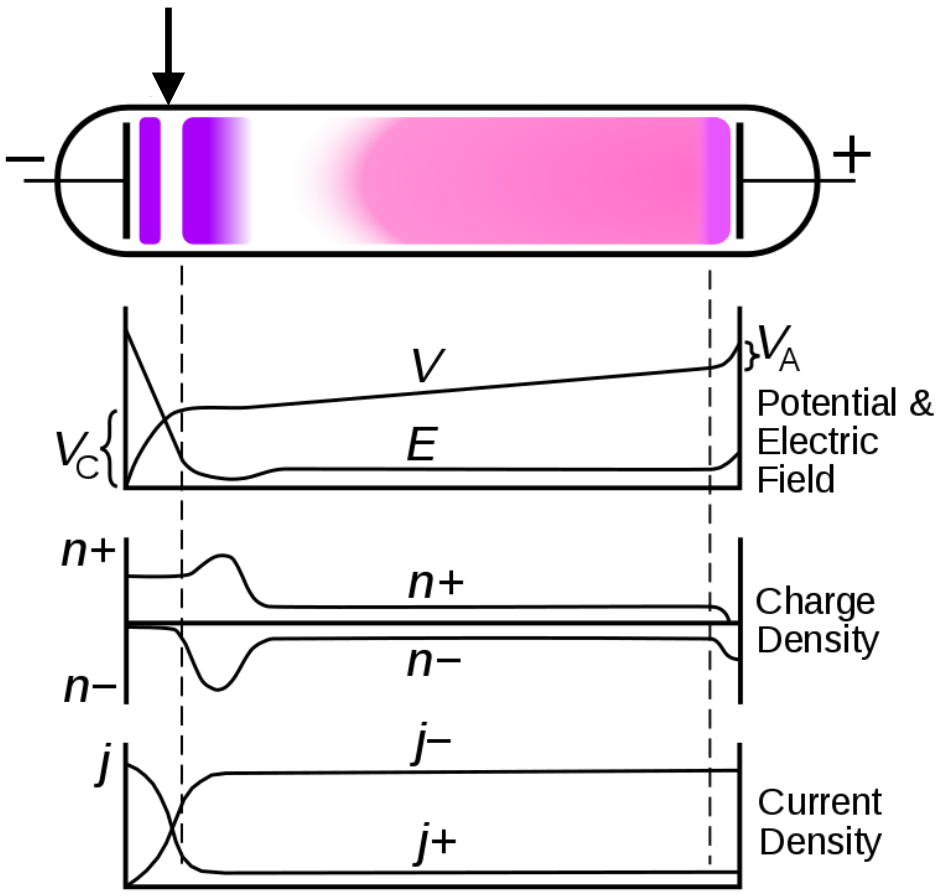
The glow discharge

- **Characteristic zones of a typical glow discharge – the cathode layer**
  - <u>Cathode dark space</u>
    aka Hittorff dark space or Crookes dark space
  - Further increase in energy of electrons from the cathode → preferentially ionisation instead of excitation → predominant separation of charge carriers (electron collision-induced ions + further electrons) instead of emission of light → large voltage drop*



*Due to the large voltage drop starting within the cathode dark space, this region and the neighbouring Aston dark space and cathode glow are also referred to as the cathode fall (voltage 'falls' towards cathode)

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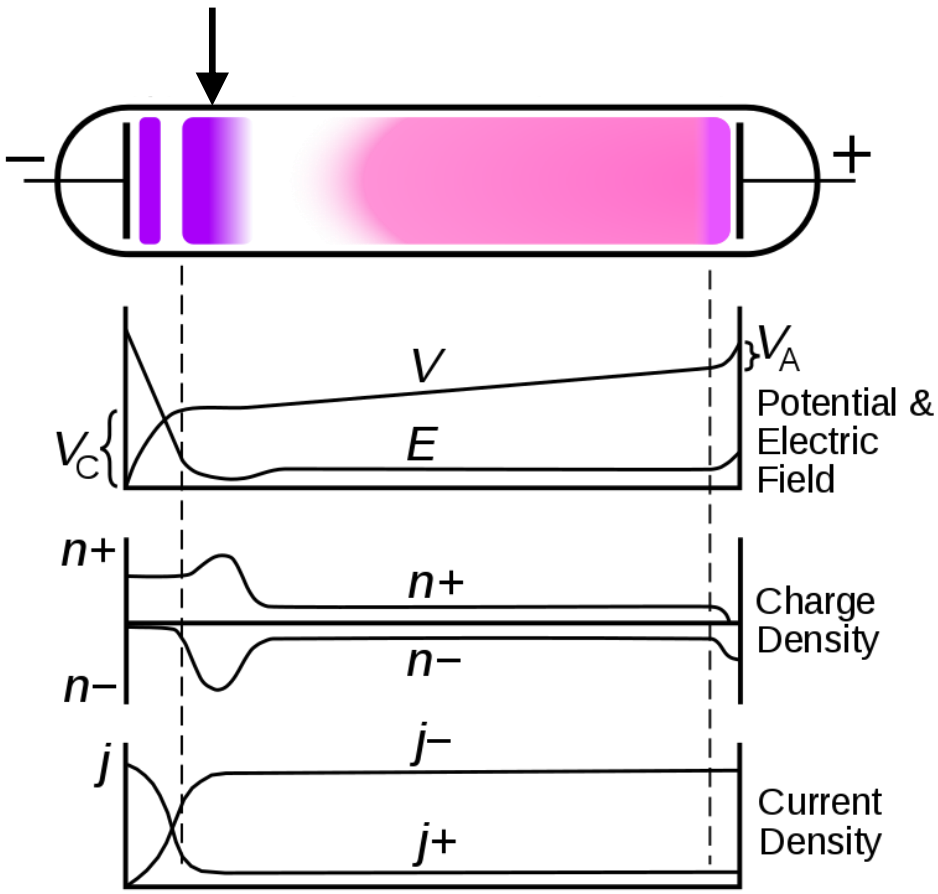
The glow discharge

- **Characteristic zones of a typical glow discharge – the cathode layer**
  - <u>Negative glow</u>
  - Ionisation within cathode dark space leads to high electron density; electrons are decelerated and finally recombine with positive ions → intense emission of light via bremsstrahlung and recombination



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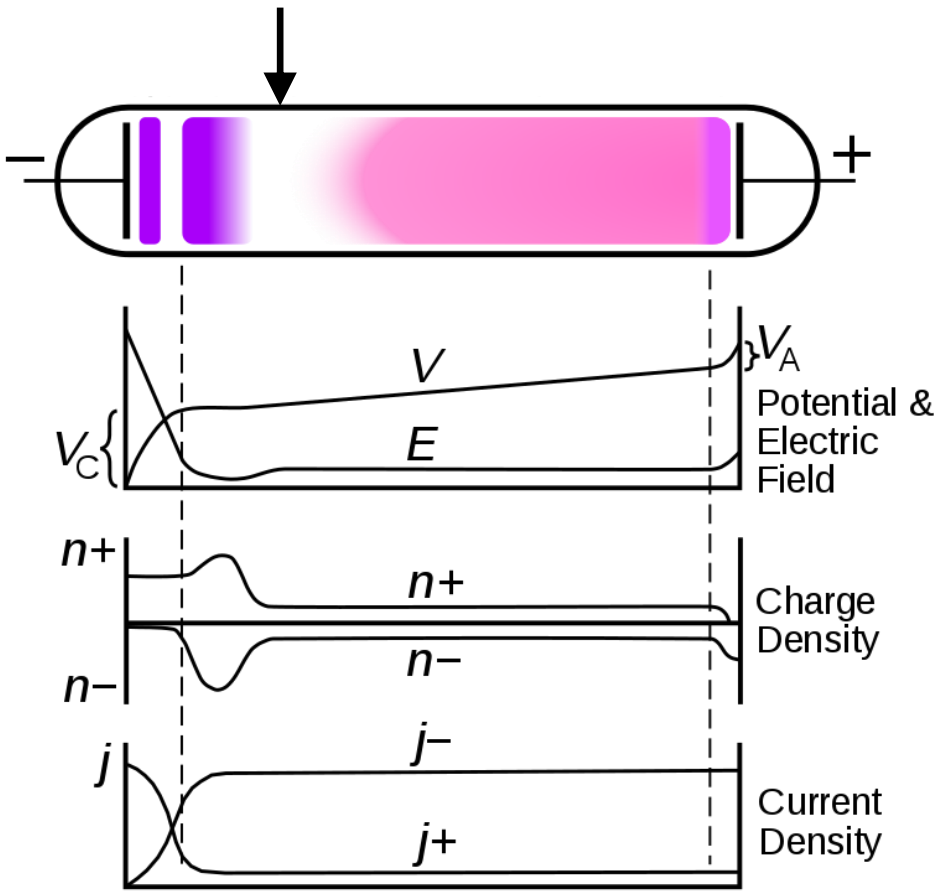
The glow discharge

- **Characteristic zones of a typical glow discharge – the cathode layer**
  - <u>Faraday dark space</u>
  - Electrons are further decelerated, remaining energy is insufficient for excitation of atoms and emission of light, respectively



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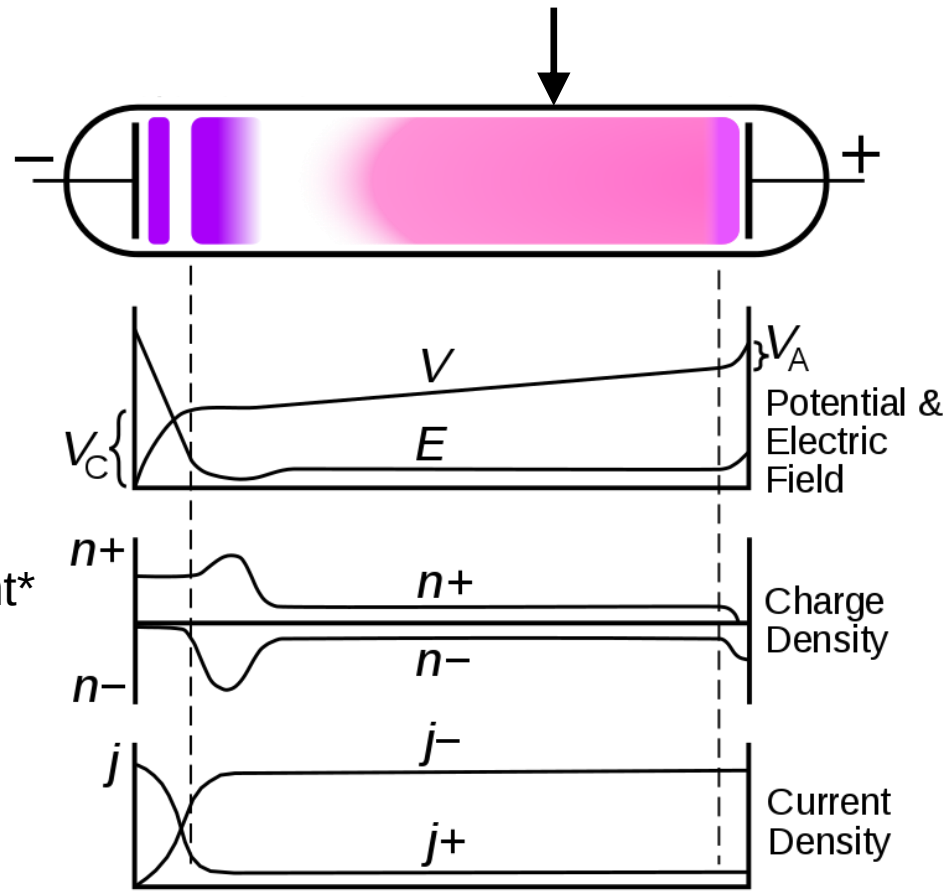
The glow discharge

- **Characteristic zones of a typical glow discharge – the anode layer**
  - <u>Positive column</u>
  - Recombination within negative glow
    → reduction in ion density
    → increase in electric field
    → acceleration of electrons up to an energy of about 2 eV
    → sufficient energy for excitation of atoms and accompanying emission of light*



*Actually, the positive column is the 'luminous element' of fluorescent tubes or high-pressure lamps. (The longer the glow discharge tube, the longer the positive column, whereas the cathode layer thickness remains almost constant.)

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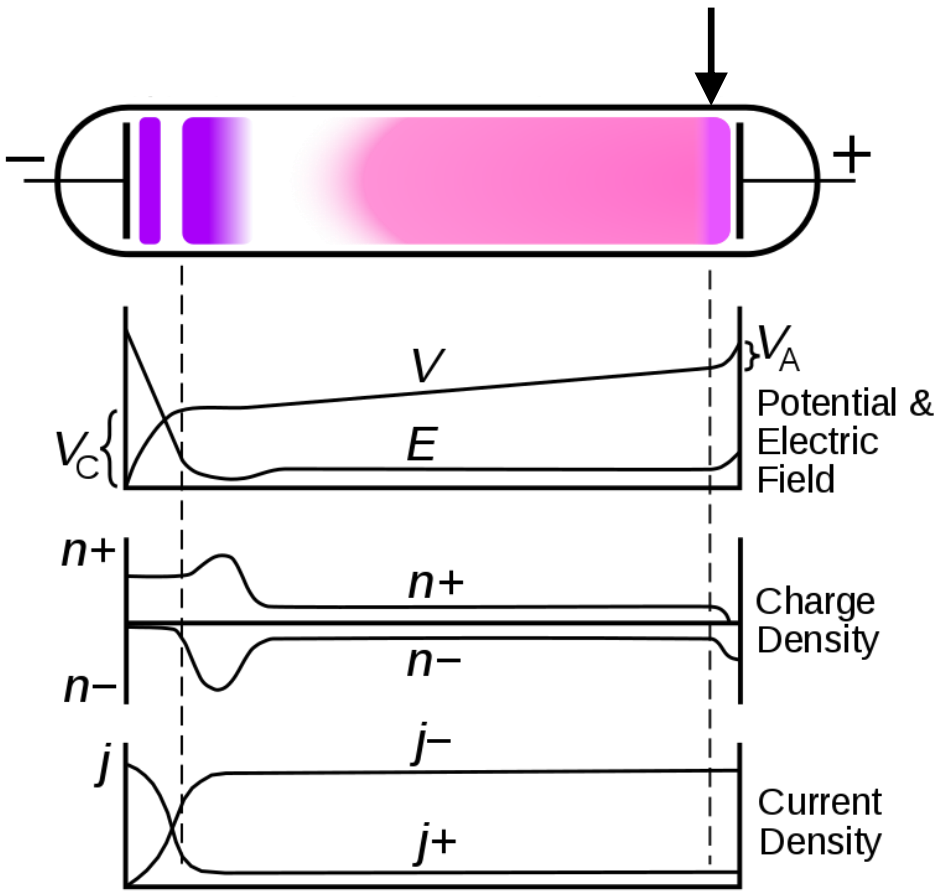
The glow discharge

- **Characteristic zones of a typical glow discharge – the anode layer**
  - <u>Anode glow</u>
  - Steep increase in electric field
    → increased number of excitation events
    → increased emission of light



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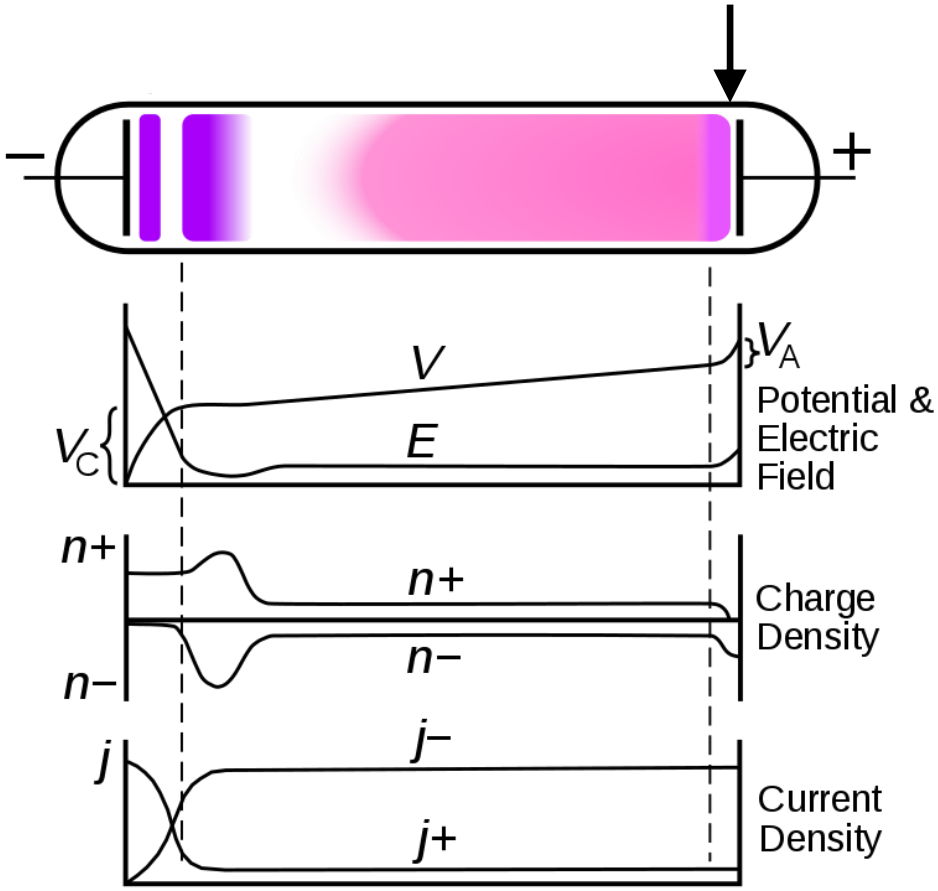
The glow discharge

- **Characteristic zones of a typical glow discharge – the anode layer**
  - <u>Anode dark space</u>
  - Reduction of electrons with sufficient energy for excitation within anode glow → no emission of light and thus dark space



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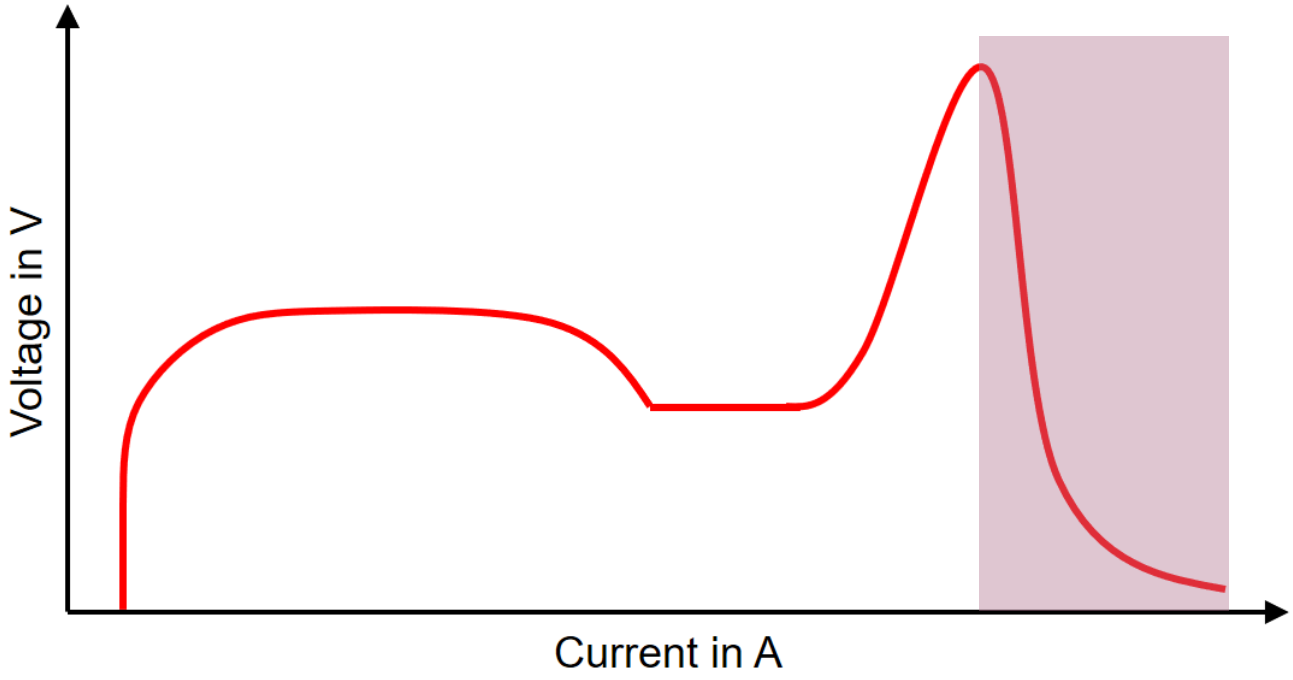
2.5 The arc discharge

Review: DC discharges

- **Arc discharge**
  - High current leads to notable heating of cathode surface
  - → Thermionic emission of further charge carriers into discharge gap



The arc discharge

- **Transition from glow discharge to arc discharge**
  - Further increase in voltage leads to transition from glow discharge to arc discharge due to an increase in current density within the plasma
    → Increased current density causes heating of cathode
    → Apart from emission of secondary electrons caused by ion impact, further electrons are removed from the electrode material via thermionic emission
    → After a certain duration, thermionic emission becomes dominant (However, heating by ion impact must compensate loss in energy/temperature of cathode caused by heat conduction or cooling by the emitted electrons.)
    → Current density of emitted electrons increases exponentially with temperature

  - Usually, arc discharges start at micro peaks on cathode surfaces, which generate high electrical fields, leading to a high ion current density towards such micro peaks

The arc discharge

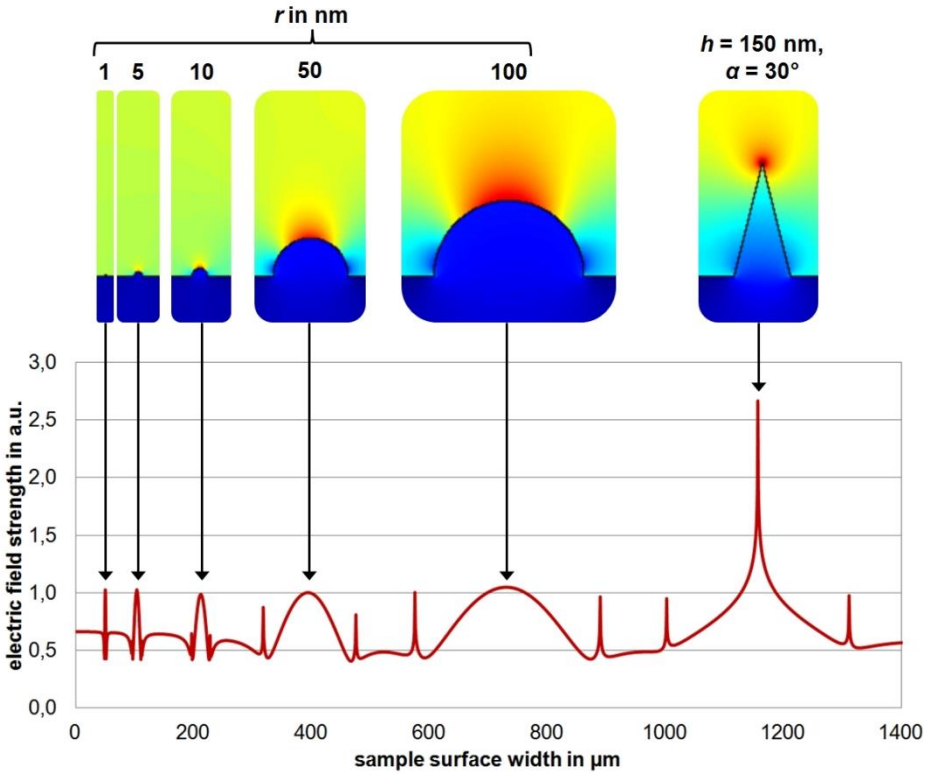
- **On the distribution of electric field strength**
  - Arc discharge usually start at micro or roughness peaks on the cathode surface where high electric fields

  → Impact on spatial structure of plasma

  → Merely the positive column and the electrode falls remain (the latter at reduced potential difference of about 10 V)

  → At low gas pressure, the plasma discharge cannot remain isothermal

  → Isothermal arc discharges are formed above a certain gas-dependent pressure (normal case)



The arc discharge

- **On the effect of cathode heating**
  - Condition for thermionic emission:
    - Increased cathode temperature (and energy, respectively), so that a sufficient amount of electrons can overcome the electric-potential barrier at the cathode surface via tunnelling from the conduction band into discharge gap
    - → Emitted electrons and heat conduction cause cooling of cathode

    - Cooling = loss in energy, which has to be balanced by energy and ion flow from plasma
      - By heating the cathode actively (usually by applying an electrical current), arc discharges can be operated at low pressure
      - Heating of the cathode can also be realised by a suitable cathode geometry, leading to local ion flow onto a small area of the cathode and its accompanying heating by ion impact

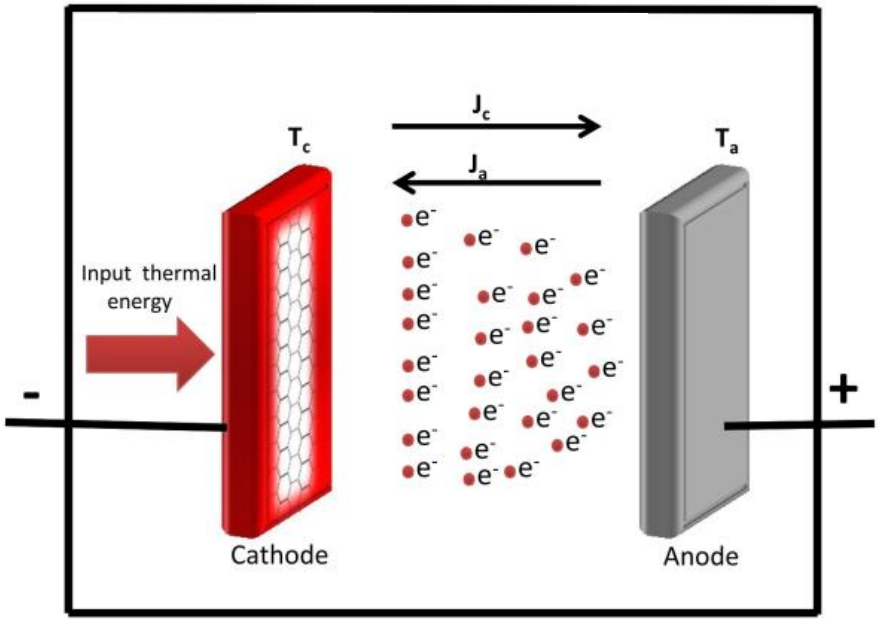
The arc discharge

- **On the effect of cathode heating**
  - Description of thermionic emission (aka Edison-Richardson effect):
  - Current density of thermionically emitted electrons is described by the Richardson equation, given by

$$J = A \cdot T^2 \cdot e^{-\left(\frac{W_e}{k_B \cdot T}\right)}$$

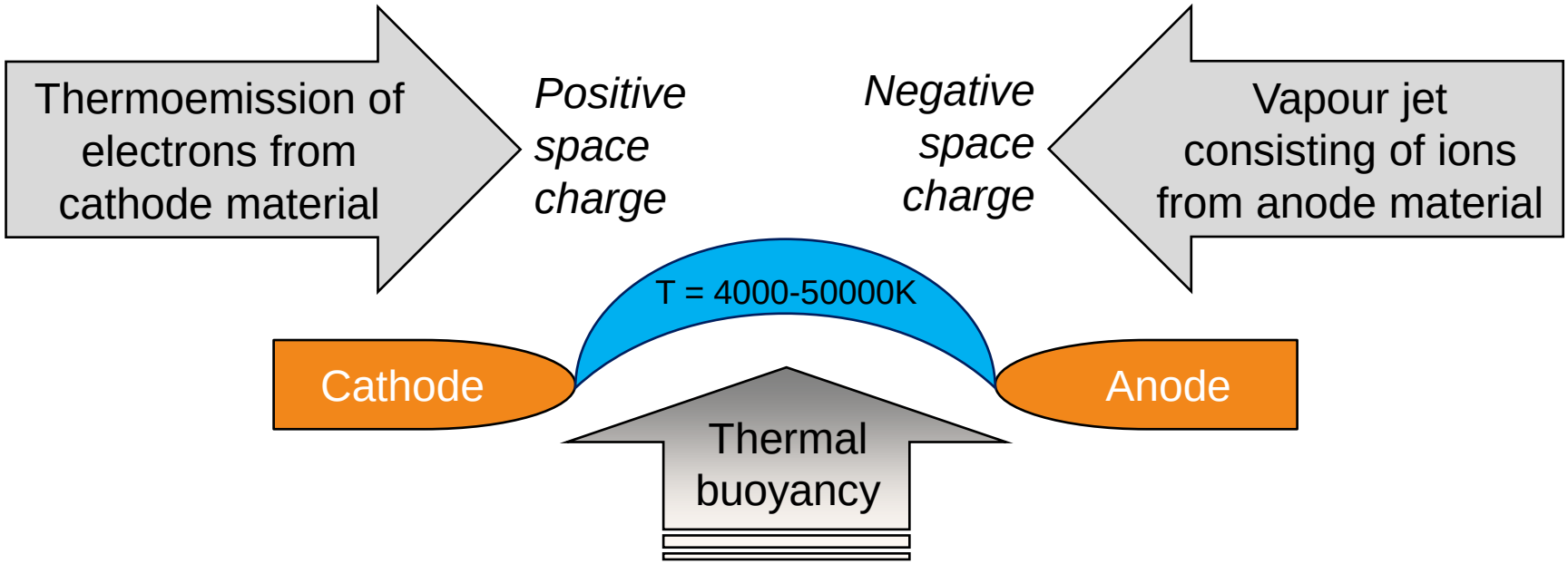
A: Richardson constant, depending on material and texture of electrode surface
T: absolute temperature
W_e: work function (here: for electrons as indicated by index e), depending on material of electrode
k_B: Boltzmann constant

For a constant area and work function, the current density exclusively depends on temperature.



The arc discharge

- **On the naming**
  - ‘Arc discharge’ or ‘arc plasma’ following the bent/arched shape of the plasma in free atmosphere, which is formed due to heating and buoyancy of the heated surrounding gas
  - Characteristics of arc discharges in atmosphere:
    - Plasma is quasi-neutral except the regions close to the electrodes
    - Dominating electron current where the required voltage is 30-100 V and the plasma temperature amounts to 1-5 eV (= 11604-58020K)



The arc discharge

- **Properties**
  - Arc discharge = thermal plasma (all particles and all degrees of freedom as for example rotation or vibration feature an almost common temperature
  - In a stationary discharge, the electron temperature T_e is, however, always some percent higher than the other temperatures (due to the higher mobility of electrons)
    - → Electrons gain higher energy from electric field (i.e., when passing through a potential difference)
    - → Electrons partially transfer kinetic energy (about 0.01% in case of elastic collision) to atoms, ions and molecules via collisions
  - Such collisions are quite rare at low pressure; here, the electrons gain higher energy and temperature of some eV whereas the gas temperature remains comparatively low.
  - At higher pressures, the electron density reaches higher values, leading to an increased number in electron-particle collisions and an accompanying increase in gas temperature and decrease in electron temperature, respectively
    → isothermal plasma

The arc discharge

- **Properties**
  - Arc plasmas at atmospheric pressure are practically in thermodynamic equilibrium = isothermal.
  - The current density in arc discharges amounts to approx. 1 kA/cm^2.
  - In case of high conductivity, the electric current merely requires a small plasma cross section.

  - The transition to an isothermal plasma comes along with a constriction of the plasma.
  - Such constriction is supported by the cathode mechanism:
    - A geometrically favoured part of the cathode (e.g. a roughness peak) is preferentially heated, causing high thermionic emission of electrons and such a high local current density

  - A thermal pulse discharge plasma is also referred to as **spark***.

**Fun fact: 'spark' = nickname for radio operators since in former times, the acoustic radio signal was generated by plasma spark discharges.*

The arc discharge

- **Characteristics**
  - The position of a thermal arc is usually static and stable, leading to a fix plasma footprint on an electrode surface.
    - However, in case of magnetic fields induced by the electric current or other external influences, such footprints move over the electrode surface
    - Motion of arcs is merely hindered by mechanical friction of the arc column with the surrounding gas

  - In case of thermionic emission, such motion starts if neighbouring areas on the cathode surface are sufficiently heated
    - Possible temperatures: 4000-50000K
      → Due to high current, plasma temperatures are notably higher than temperature of electrodes
      → Further increase in current leads to further increase in plasma temperature (application: plasma arc cutting or plasma arc welding)

https://www.fabtechexpo.com/blog/2015/03/05/selecting-plasma-arc-cutting-system

The arc discharge

- **Characteristics**
  - Non-isothermal arc plasmas at low pressure connect to the cathode at diffusively and arbitrarily distributed spots
    - Such spot footprints, so-called filaments, are non-static and move over the cathode surface.
    - The local current density is extremely high and amounts up to 10 000 A/cm^2.

The arc discharge

- **Characteristics**
  - Processes at the cathode of an arc discharge:
    - Thermionic emission (> 3000 K)
    - Field emission (MeV/cm)
    - Ion- and photon-induced emission (poor efficiency of merely 0.1%)
    - Generation of a vapour jet consisting of anode material with low ionisation energy

  - Heating-induced increased electron current form the cathode is caused by the thermo-field emission process.
  - The effective work function of the cathode material is reduced by the high electric field strength at the cathode (= Schottky effect, aka field enhanced thermionic emission)

    → $W_e = W - \Delta W$ (in Richardson equation)

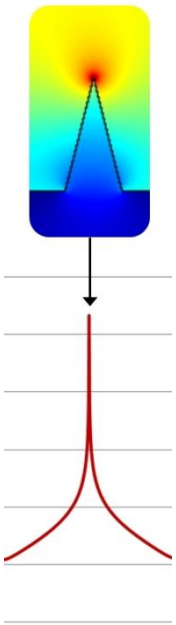
    W = nominal work function; ΔW = reduction in work function caused by electric field

  - Moreover, ions are accelerated towards the cathode and gain high velocity and kinetic energy, respectively, for removal of charge carriers via sputtering

The arc discharge

- **Characteristics**
- Extremely high electric field strength at peaks on cathode surface
  → Acceleration of ions towards peaks

- Lower losses by thermal flow at peaks
  → Higher temperature of peaks compared to rest of electrode surface



→ Peaks are preferred sources for thermionic emission of electrons

→ Arcs preferentially start at cathode peaks

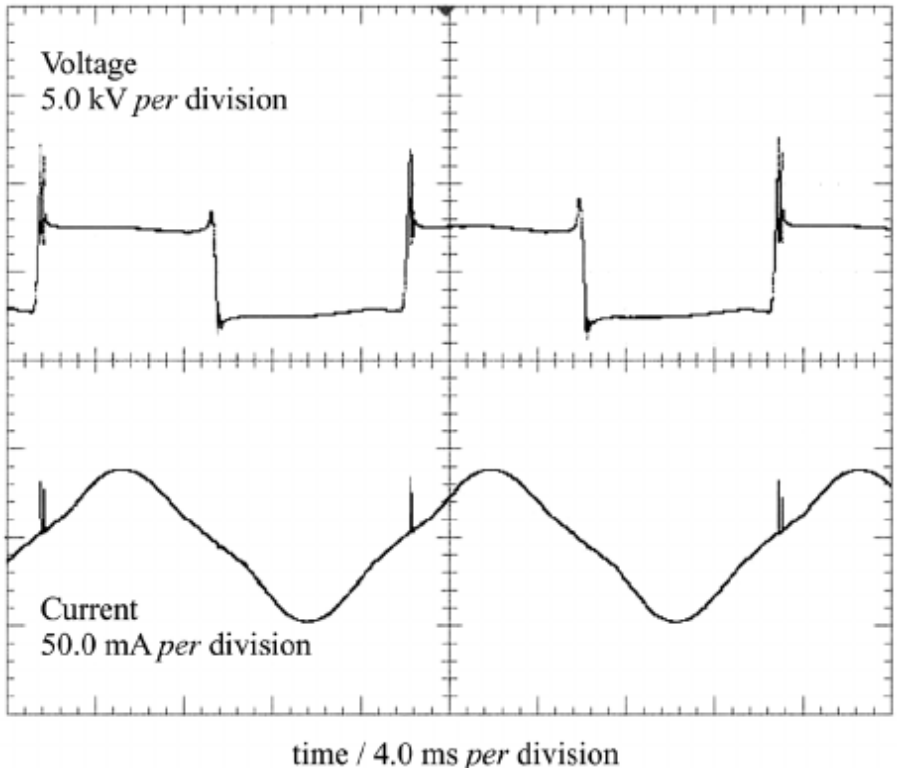
→ Microscopic filaments with high current density, i.e., about 10^8 A/cm^2

2.6 AC discharges

AC discharges

- **Differences DC-AC discharges**
  - Application of alternating current (AC) or rather: **alternating voltage**

  → Oscillating motion of electrons and ions between electrodes in intervals given by frequency or repetition rate of the electric field (note: different efficiency/velocity for electrons and ions, see ‘drift-diffusion-approximation’ in 1.2.5)



DOI:10.1590/S0103-50532011000900008

AC discharges

- **Differences DC-AC discharges**
  - At low frequency (< some kHz) and in case of high amplitudes of the oscillating motion with respect to the discharge gap, an initial cathode mechanism becomes crucial:
    - → Electrons form the cathode material are transferred into the plasma
    - → Formation of a glow discharge or arc discharge, depending on the type of gas (1^{rst} Townsend coefficient), gas density (pressure), current density, and cathode material (2nd Townsend coefficient)
    - → Character of discharge changes when increasing frequency of alternating voltage
    - → Cathode mechanism increasingly looses in importance

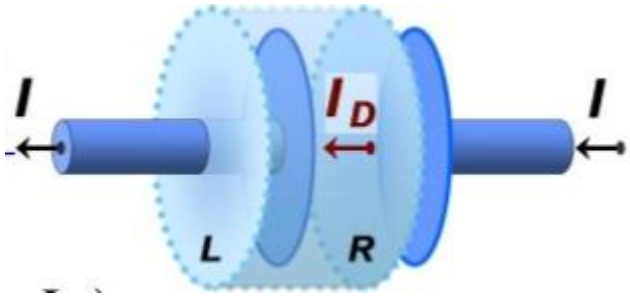
AC discharges

- **Differences DC-AC discharges**
  - For DC discharges, the power coupled into the plasma depends on the ohmic DC-current flowing between the electrodes

  - For AC discharges, notably higher charge carriers densities are achieved.
  - The plasma is then essentially heated by the displacement current* whereas the ohmic current plays a marginal role.

*Displacement current I_D = part of the electric current, which follows from the temporal change in electrical flow. It results from the orientation and formation of electrical dipoles.

Example: discharging of capacitors:



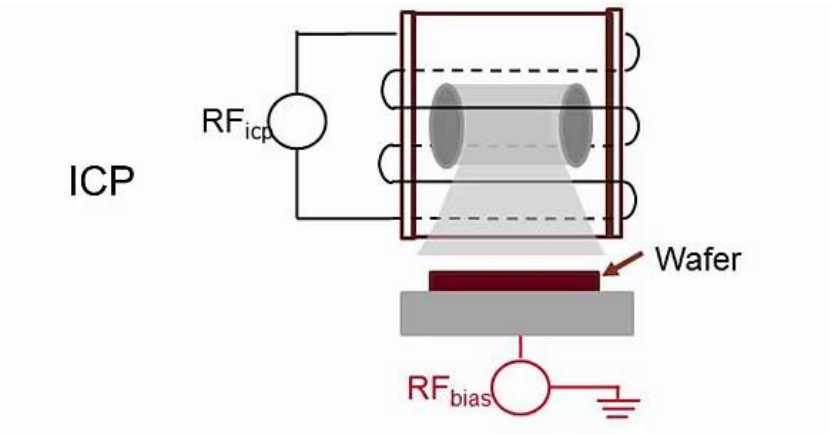
$$I_D = \frac{dQ_{\text{inside}}}{dt}$$

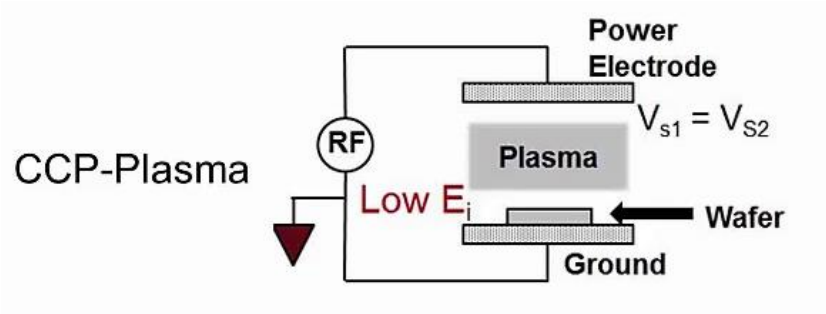
AC discharges

- **Types of excitation**
  - For the technical realisation of an AC-excitation, two approaches can be chosen for the coupling of energy into the plasma:

    1. Coupling of energy via magnetic field component *B*
       = Inductively coupled plasma (ICP)

    2. Coupling of energy via the electrical field *E*
       = Capacitively coupled plasma (CCP)





AC discharges

- **Types of excitation**
  - Coupling of energy via magnetic field component *B*
    = Inductively coupled plasma (ICP)

  - This type of coupling is also referred to as ‘inductive heating’.

  - Here, the plasma is located within a coil and an alternating current *J* flows though this coil.

  → The temporal periodically magnetic field $H(t)$, which is formed within the coil, induces a periodically alternating electrical voltage and a rotating electrical field, respectively

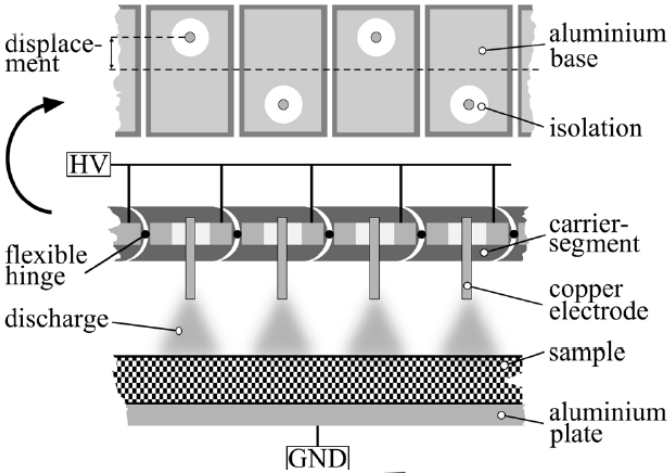
  → Electrons are accelerated within the formed rotating electrical field and generate a circular electron current J_e, which is delayed by a phase π with respect to the current *J*

AC discharges

- **Types of excitation**
  - Coupling of energy via the electrical field *E*
    = Capacitively coupled plasma (CCP)

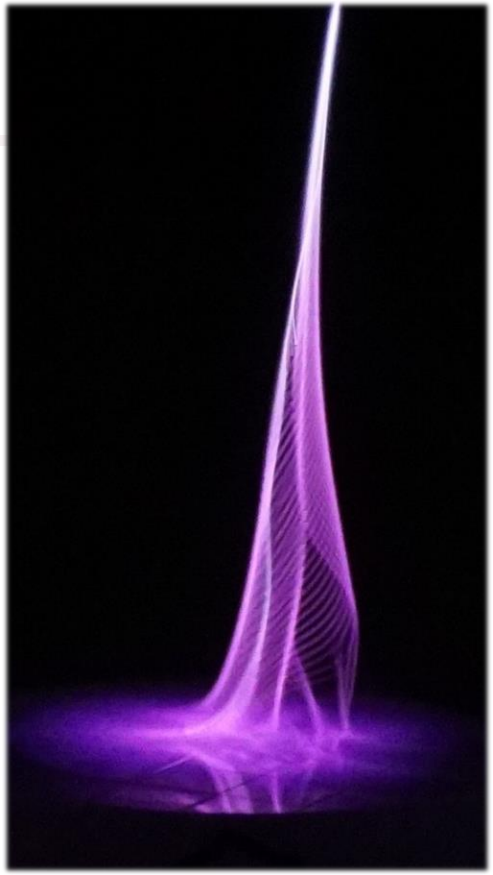
  - This type of discharge is also referred to as ‘silent discharge‘ or ‘barrier discharge’ (in special cases see c) below, also ‘dielectric barrier discharge’)

  - The high frequency can be
    a) applied directly to the electrode,
    b) fed via a capacitor, or
    c) fed via an electrically isolating layer, which is located between the electrodes and the plasma. This layer represents the barrier and acts as a capacitor, limiting the current density and stabilising the barrier discharge
    → dielectric barrier discharge (DBD)



2.7 Dielectric barrier discharges (DBD)

Dielectric barrier discharges



- **Overview**
  - Main features of DBD plasmas:
    - Non-equilibrium plasma ($T_i \neq T_e$)
    - Usually operated at atmospheric pressure
    - aka ‘cold’ plasmas due to low gas temperature
    - Usually consisting of single filaments

  - Typically driven at
    - $U \approx 10$ kV,
    - $f \approx 10$ kHz, and
    - $P < 1$ W

  - Used working gases
    (depending on desired target effect, i.e. either plasma-chemical or plasma-physical impact on work piece or mixture of both)
    - Air
    - Argon
    - Hydrogenous gas mixtures (e.g. forming gas or varigon)
    - etc.

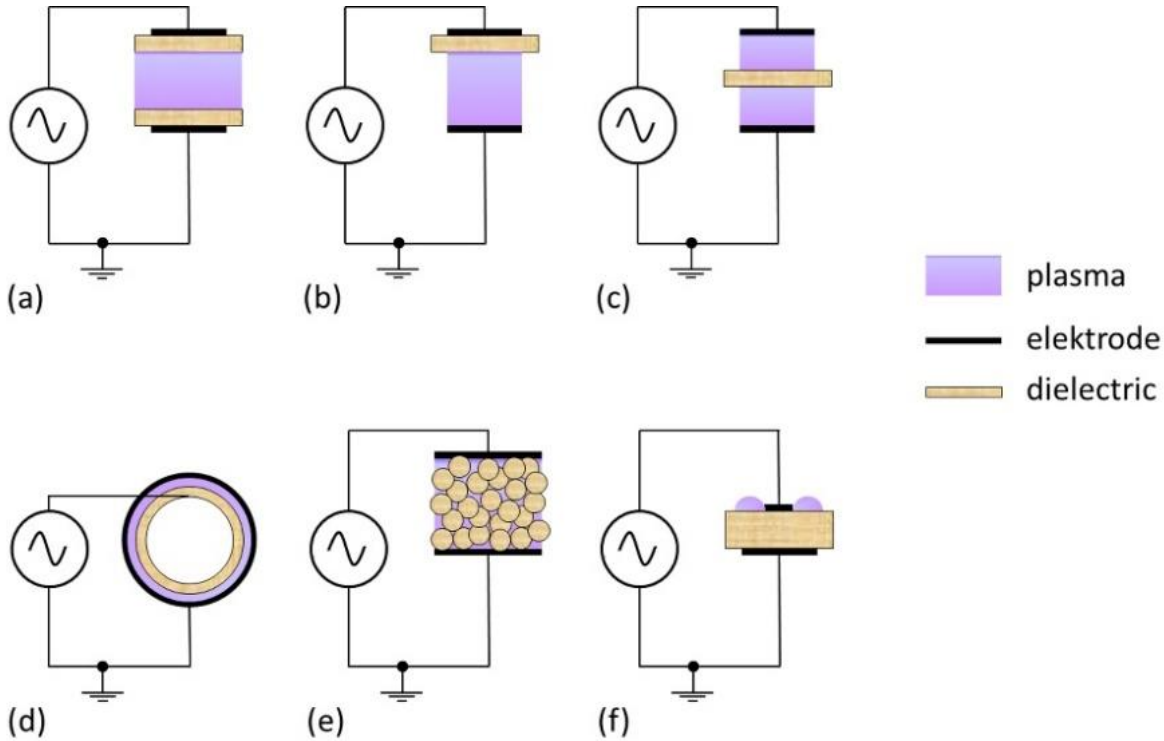
Dielectric barrier discharges

- **Overview**
  - Typical physical parameters of single filaments of dielectric barrier discharges:

Parameter	Value
filament radius	$\approx 10^{-4}$ m
max. current	0.1 A
current density	$10^{6} \ldots 10^{7}$ A/m^{2}
electron density	$10^{20} \ldots 10^{21}$ 1/m^{3}
mean electron energy	1…10 eV
temperature	$\approx$ 300 K
total charge	$10^{-10} \ldots 10^{-9}$ C
duration	$10^{-9} \ldots 10^{-8}$ s

Dielectric barrier discharges

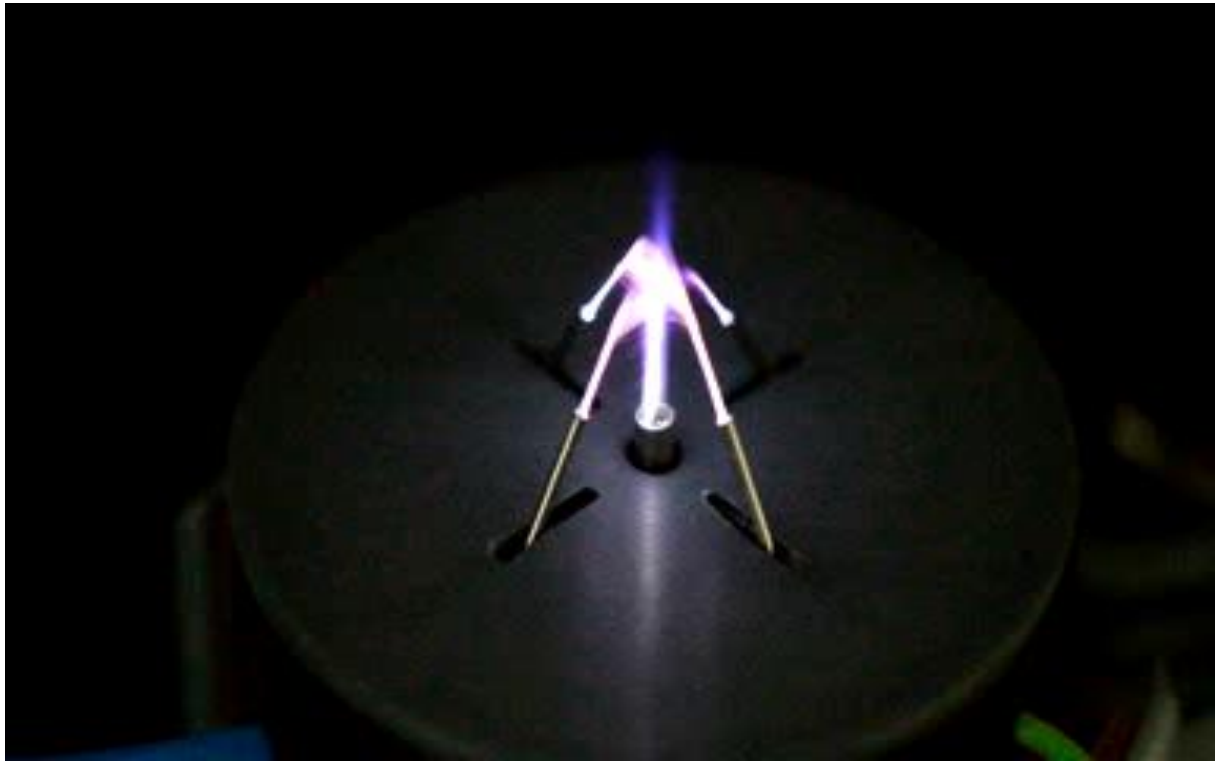
- **Characteristics: Dielectric separation of at least one electrode**
  → Marginal electric current (some nA - µA)
  → Moderate heating



a)-c) planar discharge arrangement; d) cylindrical discharge arrangement; e) „Packed-Bed“ discharge; f) surface discharge arrangement

Dielectric barrier discharges

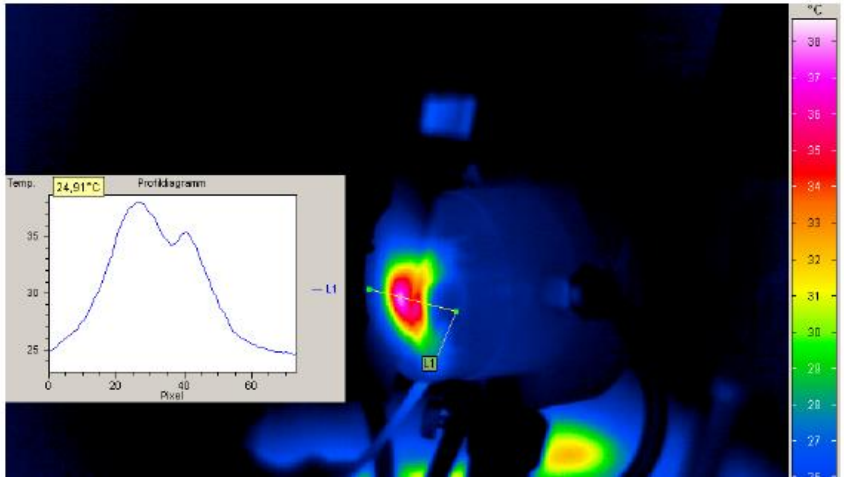
- **Characteristic ‘moderate heating’**
  - Marginal current → marginal heating → ‘cold’ plasma



Dielectric barrier discharges

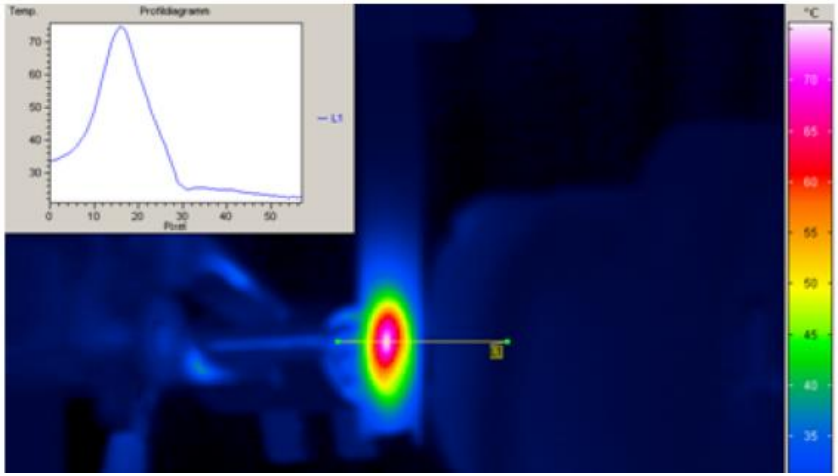
- **Characteristic ‘moderate heating’**
  - Marginal current → marginal heating → ‘cold’ plasma

indirect (jet):



T_{max} = 38°C

direct:

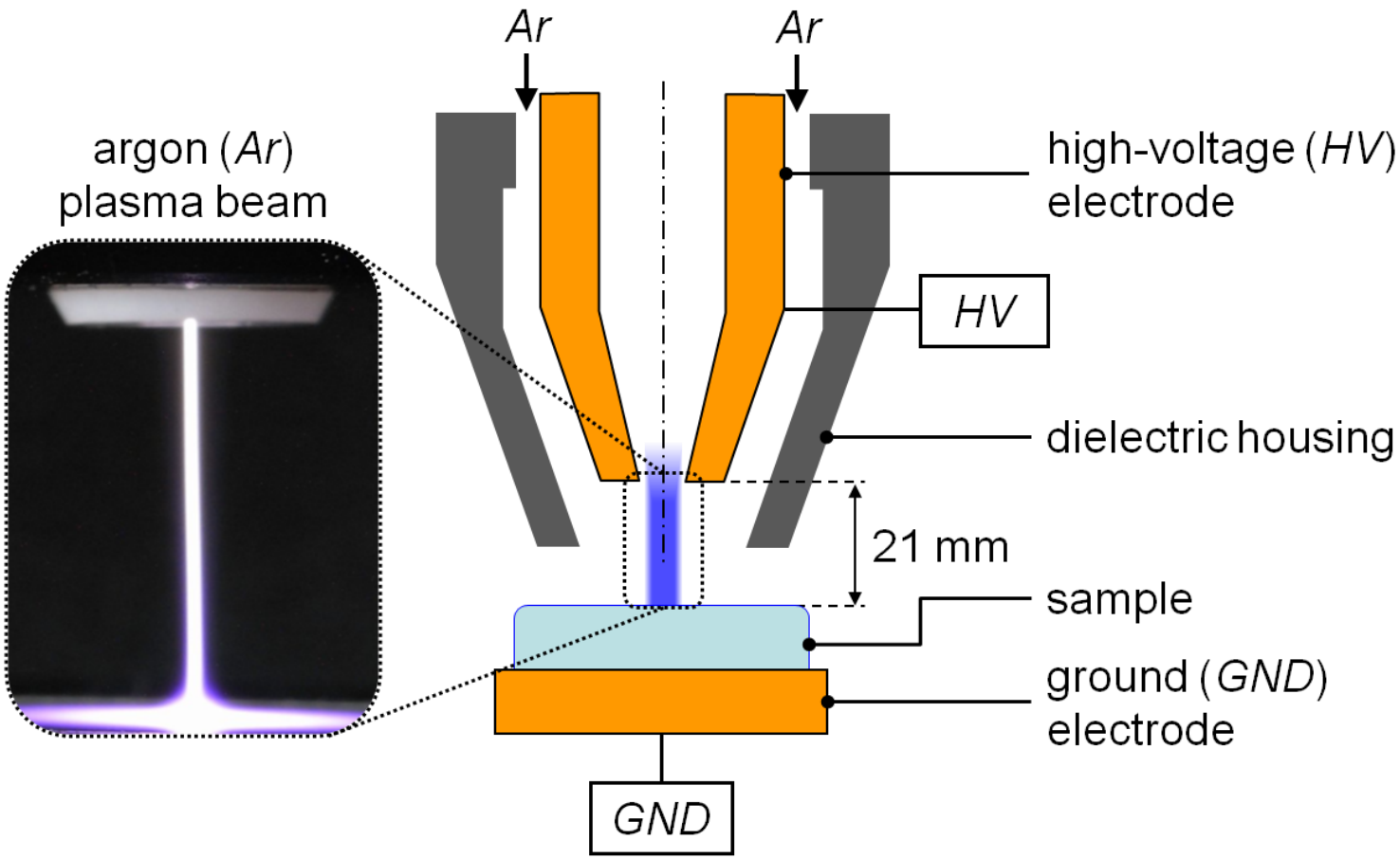


T_{max} = 88°C

- → suitable for the treatment of temperature-sensitive materials

Dielectric barrier discharges

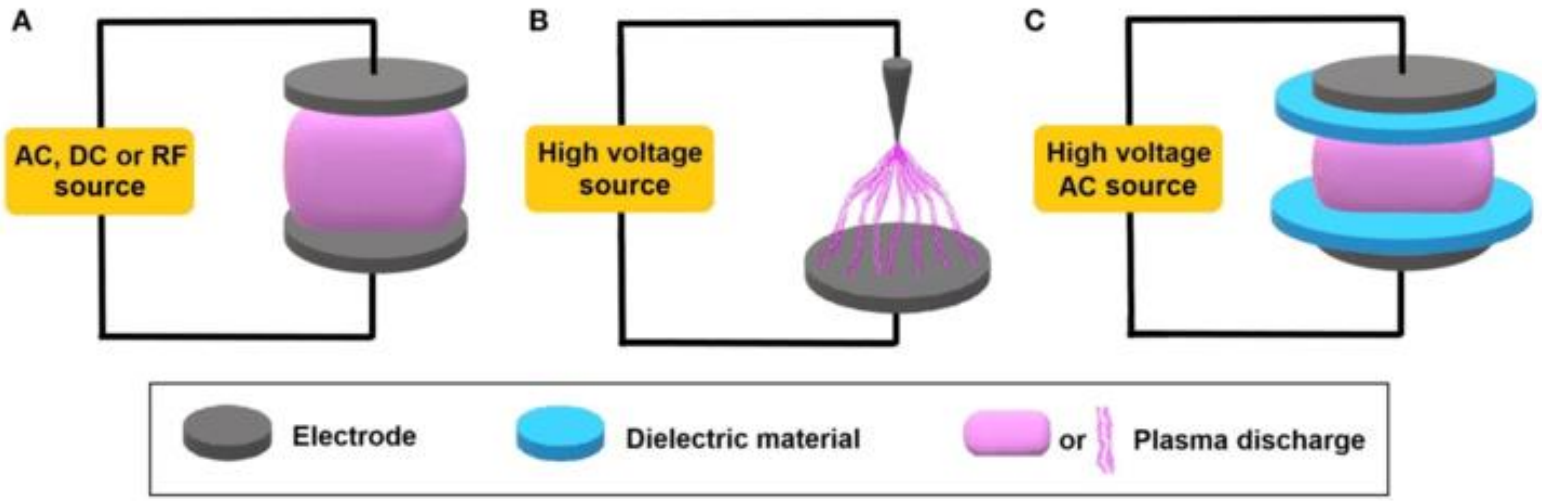
- **Example: Direct DBD plasma source**



(Possible) applications: See following chapters…

Dielectric barrier discharges

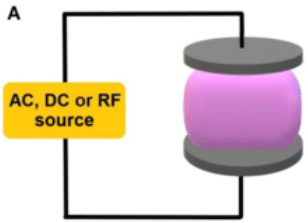
- **Worked example 2.4**
  - Suggest some technical applications for the three different types of plasma shown below.



DOI:10.3389/fmech.2020.00042

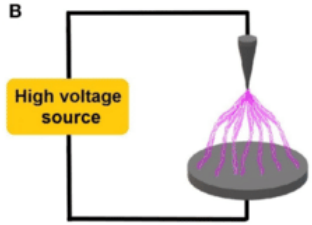
Dielectric barrier discharges

- **Worked example 2.4 – Answer**
  - A = hot and homogeneous discharge



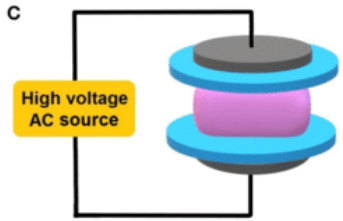
    - Applications: welding and cutting (classical circuit of plasma flame cutter)
    - Device: ignition plug

  - B = hot, filamentary, and inhomogeneous (plasma streamers with high energy density)



    - Cell perforation
    - Surface structuring
    - etc.

  - C = cold and homogenous



    - Gentle surface activation and functionalisation (of temperature-sensitive materials, e.g. plastics or papers)
    - Plasma medicine
    - Surface cleaning
    - etc.

Dielectric barrier discharges

???

- **Exercise 2.8**
  - Calculate the optimum discharge gap thickness of a dielectric barrier discharge plasma at atmospheric pressure where the discharge gap is filled with helium and the electrodes are covered with sodium glass plates as dielectric. For solving this task, the following table was found in literature:

Table 1 Paschen coefficients for several commonly used gases (taken from [4]).

gas	electrode	A $[Pa^{-1}m^{-1}]$	B $[VPa^{-1}m^{-1}]$	γ
He	sodium - glass	2.10	25.50	0.263
N_2 (air)	platinum	10.95	273.78	0.025
Ar	tungsten	10.20	176.27	0.095

[4] J. Cobine, ”Gaseous conductors”, (Dover, New York, 1958).

Dielectric barrier discharges

???

- **Exercise 2.9**
  - An atmospheric pressure plasma in local thermal equilibrium should be used for surface activation of a plane work piece made of PVC, where the operating temperature of PVC is 60°C (above this temperature, the work piece made of PVC deforms). The plasma is operated at a high voltage of 15 kV; the discharge gap thickness is 3 cm.

  - Can such a plasma be used for the described application?